INCH-POUND

MIL-M-28787/438
26 November 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

BREAKPOINT, RCMI MODULE, KEY CODE GTC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Remote Control and Maintenance (RCMI) breakpoint module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 31.032 failures per million operating hours for class I or 124.714 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be GTC.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
B1-B8	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	Q	---------- Module shall pass all remaining tests ----------						
D1-D8	Ibd	V	S	0.0	0.005	0.0	0.005	0.0	0.005	
D9	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	
D10	Ibd	V	S	0.0	3.01	0.0	3.01	0.0	3.01	
D11,D12	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	
D13	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	
D14-D34	Ibd	V	S	0.0	0.005	0.0	0.005	0.0	0.005	
D35	Ibd	V	S	0.0	0.13	0.0	0.13	0.0	0.13	
D36-D38	Ibd	V	S	0.0	0.005	0.0	0.005	0.0	0.005	
D39-D41	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	
D42	Ibd	V	S	0.0	1.01	0.0	1.01	0.0	1.01	
D43-D45	Ibd	V	S	0.0	1.02	0.0	1.02	0.0	1.02	
D46	Ibd	V	S	0.0	0.005	0.0	0.005	0.0	0.005	
D47	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	
D48	Ibd	V	S	0.0	0.105	0.0	0.105	0.0	0.105	
D49	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	
D50,D51	Ibd	V	S	0.0	0.005	0.0	0.005	0.0	0.005	
D52	Ibd	V	S	0.0	1.31	0.0	1.31	0.0	1.31	
D53	Ibd	V	S	-0.22	0.23	-0.22	0.23	-0.22	0.23	
D54	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	
D55,D56	Ibd	V	S	0.0	0.005	0.0	0.005	0.0	0.005	
D57	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	
D58-D64	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	
D65,D66	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	
D67	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	
E1-E8	Ios	VI	S	-240.0	-40.0	-240.0	-40.0	-240.0	-40.0	
E9-E16	Ios	VI	S	-250.0	-30.0	-250.0	-30.0	-250.0	-30.0	
E17-E19	Ios	VI	S	-240.0	-60.0	-240.0	-60.0	-240.0	-60.0	
E20	Ios	VI	S	-250.0	-30.0	-250.0	-30.0	-250.0	-30.0	

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F1-F8	Ith	VII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	
F9-F14	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	
F15	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	
F16	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	
F17	Ith	VII	X	-0.76	0.005	-0.76	0.005	-0.76	0.005	
F18	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	
F19	Ith	VII	X	-0.51	0.051	-0.51	0.051	-0.51	0.051	
F20	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	
F21	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	
F22-F37	Ith	VII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	
F38	Ith	VII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	
F39	Ith	VII	X	-0.515	0.065	-0.515	0.065	-0.515	0.065	
F40-F44	Ith	VII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	
F45	Ith	VII	X	-0.63	0.05	-0.63	0.05	-0.63	0.05	
F46-F48	Ith	VII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	
F49,F50	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	
F51	Ith	VII	X	-0.255	0.03	-0.255	0.03	-0.255	0.03	
F52	Ith	VII	X	-0.265	0.04	-0.265	0.04	-0.265	0.04	
F53	Ith	VII	X	-0.27	0.045	-0.27	0.045	-0.27	0.045	
F54	Ith	VII	X	-0.275	0.05	-0.275	0.05	-0.275	0.05	
F55	Ith	VII	X	-0.27	0.045	-0.27	0.045	-0.27	0.045	
F56	Ith	VII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	
F57	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	
F58	Ith	VII	X	-0.605	0.025	-0.605	0.025	-0.605	0.025	
F59	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	
F60,F61	Ith	VII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	
F62	Ith	VII	X	-2.86	0.1	-2.86	0.1	-2.86	0.1	
F63	Ith	VII	X	-10.4	-5.52	-10.4	-5.52	-10.4	-5.52	
F64	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	
F65,F66	Ith	VII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	
F67	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	
F68-F74	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	
F75,F76	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	
F77	Ith	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G1-G8	I11	VIII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	%
G9-G14	I11	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	%
G15	I11	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	%
G16	I11	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	%
G17	I11	VIII	X	-0.76	0.085	-0.76	0.085	-0.76	0.085	%
G18	I11	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	%
G19	I11	VIII	X	-0.51	0.051	-0.51	0.051	-0.51	0.051	%
G20	I11	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	%
G21	I11	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	%
G22-G37	I11	VIII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	%
G38	I11	VIII	X	-0.015	0.015	-0.015	0.015	-0.015	0.015	%
G39	I11	VIII	X	-0.515	0.065	-0.515	0.065	-0.515	0.065	%
G40-G44	I11	VIII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	%
G45	I11	VIII	X	-0.63	0.05	-0.63	0.05	-0.63	0.05	%
G46-G48	I11	VIII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	%
G49,G50	I11	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	%
G51	I11	VIII	X	-0.255	0.03	-0.255	0.03	-0.255	0.03	%
G52	I11	VIII	X	-0.265	0.04	-0.265	0.04	-0.265	0.04	%
G53	I11	VIII	X	-0.27	0.045	-0.27	0.045	-0.27	0.045	%
G54	I11	VIII	X	-0.275	0.05	-0.275	0.05	-0.275	0.05	%
G55	I11	VIII	X	-0.27	0.045	-0.27	0.045	-0.27	0.045	%
G56	I11	VIII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	%
G57	I11	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	%
G58	I11	VIII	X	-0.605	0.025	-0.605	0.025	-0.605	0.025	%
G59	I11	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	%
G60,G61	I11	VIII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	%
G62	I11	VIII	X	-2.86	0.1	-2.86	0.1	-2.86	0.1	%
G63	I11	VIII	X	-10.4	-5.52	-10.4	-5.52	-10.4	-5.52	%
G64	I11	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	%
G65,G66	I11	VIII	X	-0.005	0.005	-0.005	0.005	-0.005	0.005	%
G67	I11	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	%
G68-G74	I11	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	%
G75,G76	I11	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	%

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G77	Iii	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
H1-H20	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J24	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1	Iozh	XI	X	-0.01	0.01	-0.01	0.01	-0.01	0.01	mA
K2	Iozh	XI	X	-0.25	0.25	-0.25	0.25	-0.25	0.25	mA
K3	Iozh	XI	X	-0.5	0.5	-0.5	0.5	-0.5	0.5	mA
K4	Iozh	XI	X	-0.25	0.25	-0.25	0.25	-0.25	0.25	mA
K5	Iozl	XI	X	-0.01	0.01	-0.01	0.01	-0.01	0.01	mA
L1-L8	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L9-L12	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L13-L36	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L37-L39	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L40	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L41	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L42	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L43	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L44,L45	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L46	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L47,L48	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L49-L58	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L59	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M	PSFC	XIII	S	6.3	11.7	6.3	11.7	6.3	11.7	uF
N	Ipo	XIV	X	600	1633	600	1633	600	1633	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ∓ 5°C		Class II 4/ -55 ± 5°C 85 ∓ 5°C		Units
				Min	Max	Min	Max	Min	Max	
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R2-R8	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R9,R10	C1/D	XVIII	Q	6	125	6	125	6	125	pF
R11,R12	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R13	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R14-R22	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R23	C1/D	XVIII	Q	2	100	2	100	2	100	pF
R24	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R25,R26	C1/D	XVIII	Q	2	100	2	100	2	100	pF
R27	C1/D	XVIII	Q	2	125	2	125	2	125	pF
R28,R29	C1/D	XVIII	Q	2	100	2	100	2	100	pF
R30-R32	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R33,R34	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R35	C1/D	XVIII	Q	6	175	6	175	6	175	pF
R36-R38	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R39	C1/D	XVIII	Q	2	150	2	150	2	150	pF
R40	C1/D	XVIII	Q	2	100	2	100	2	100	pF
R41	C1/D	XVIII	Q	4	150	4	150	4	150	pF
R42	C1/D	XVIII	Q	2	100	2	100	2	100	pF
R43	C1/D	XVIII	Q	2	125	2	125	2	125	pF
R44	C1/D	XVIII	Q	2	150	2	150	2	150	pF
R45	C1/D	XVIII	Q	2	125	2	125	2	125	pF
R46	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R47	C1/D	XVIII	Q	2	125	2	125	2	125	pF
R48,R49	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R50,R51	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R52	C1/D	XVIII	Q	6	217	6	217	6	217	pF
R53	C1/D	XVIII	Q	4	75	4	75	4	75	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R54,R55	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R56	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R57-R63	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R64,R65	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R66-R74	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R75	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R76	C1/o	XVIII	Q	8	200	8	200	8	200	pF
R77	C1/o	XVIII	Q	4	100	4	100	4	100	pF
T	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/438-1 for class I and M28787/438-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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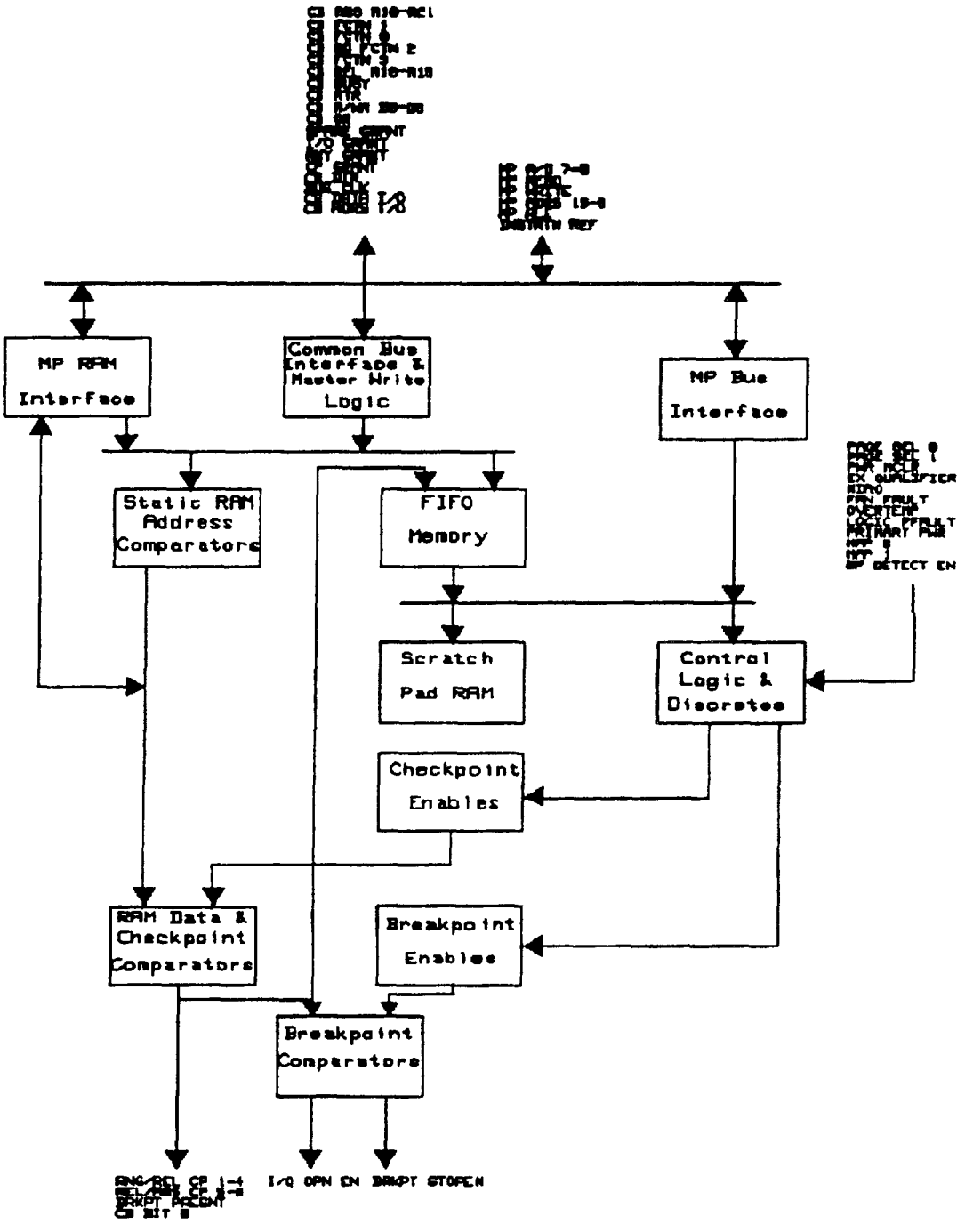


FIGURE 1. GTC functional block diagram.

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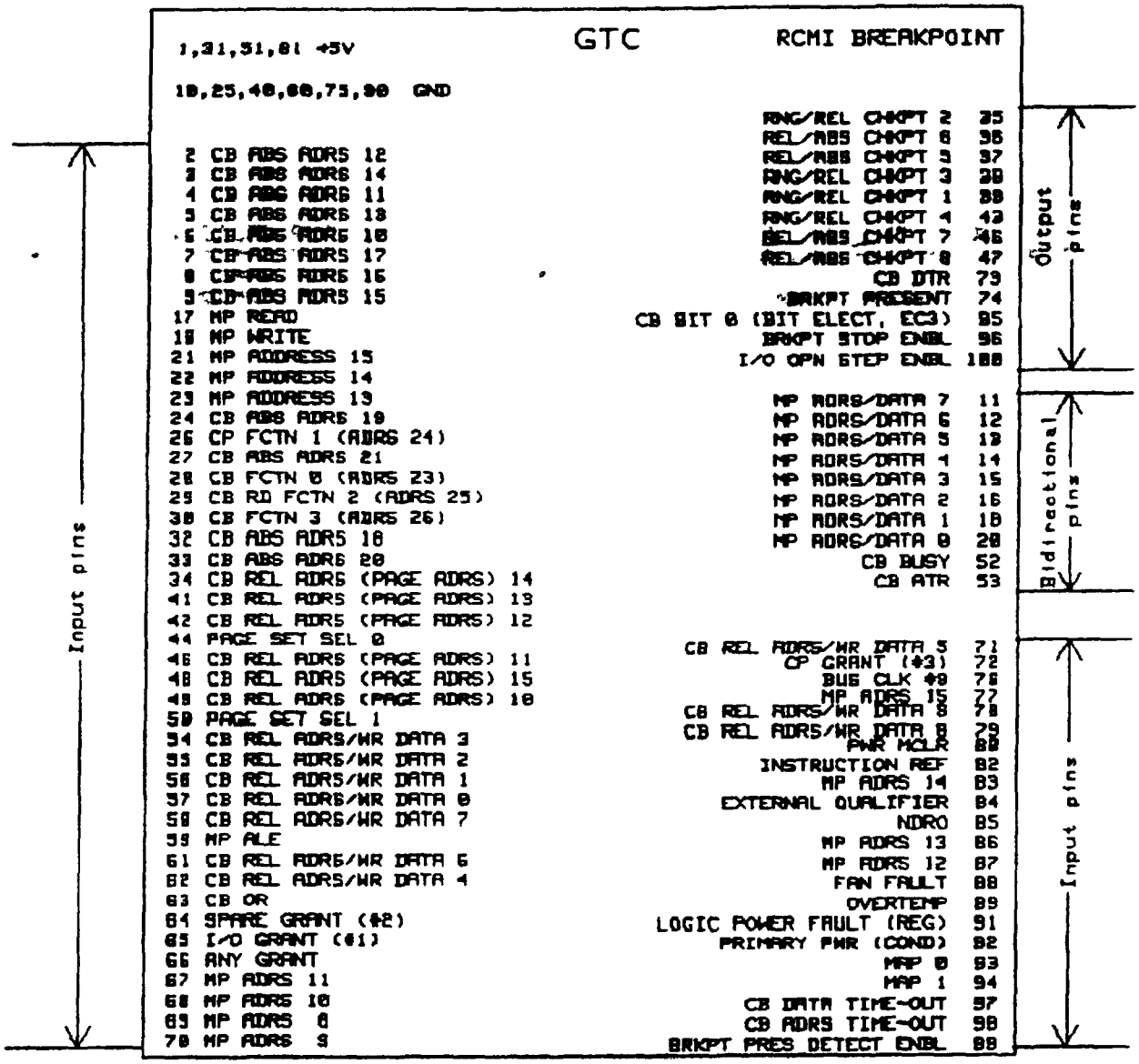


FIGURE 2. GTC pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

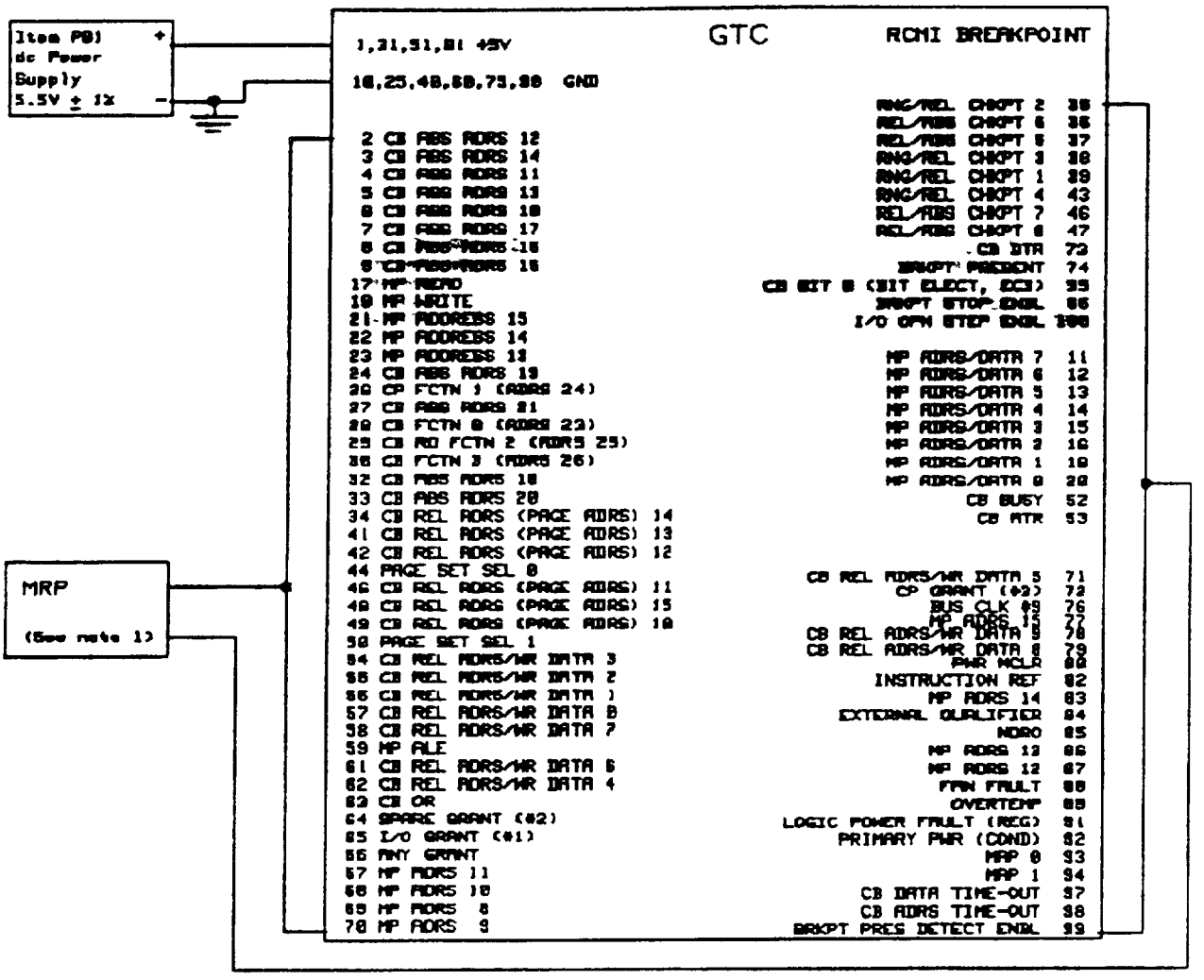
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. GTC operating temperature and component temperature test circuit.

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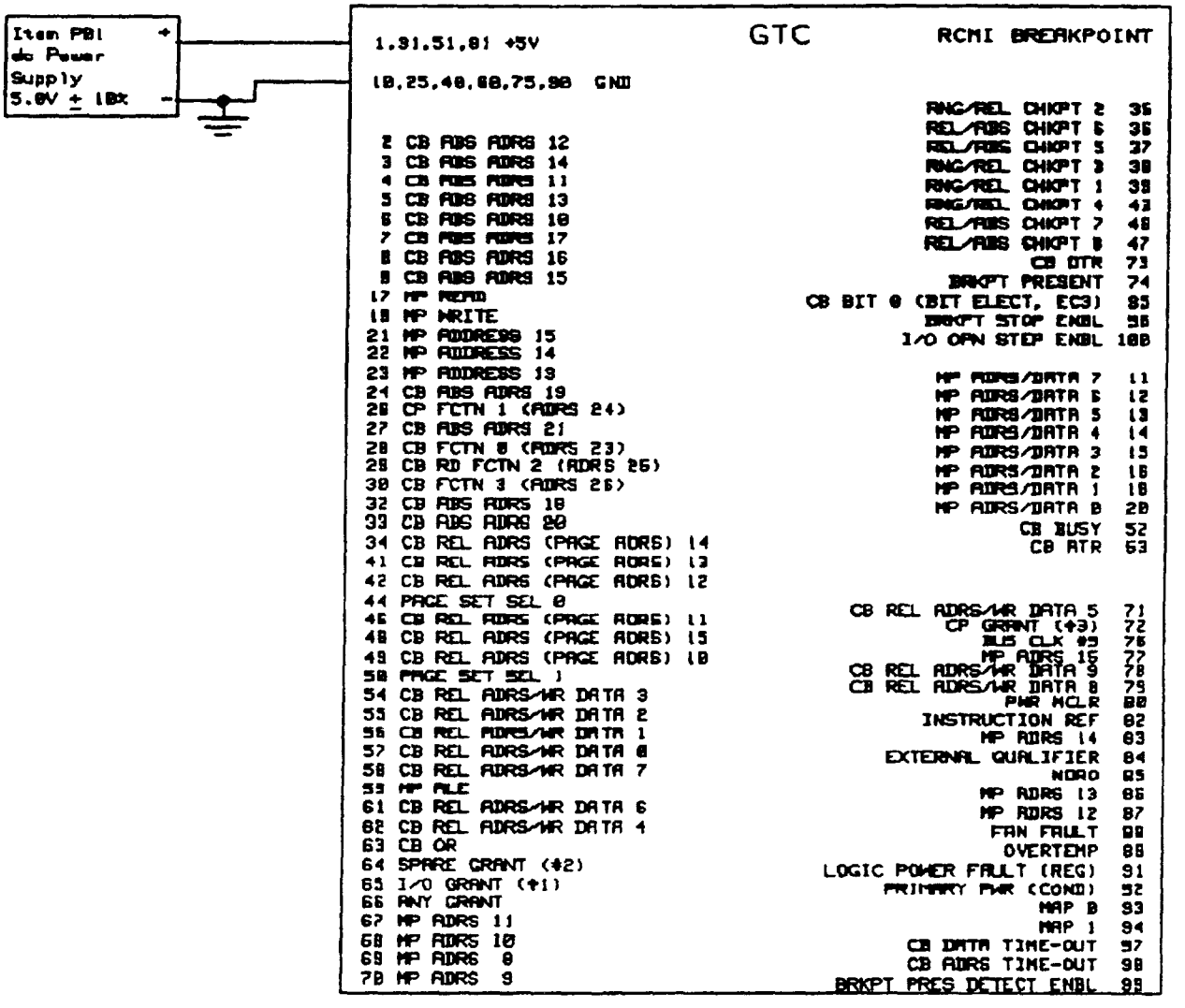


FIGURE 4. GTC vibration test circuit.

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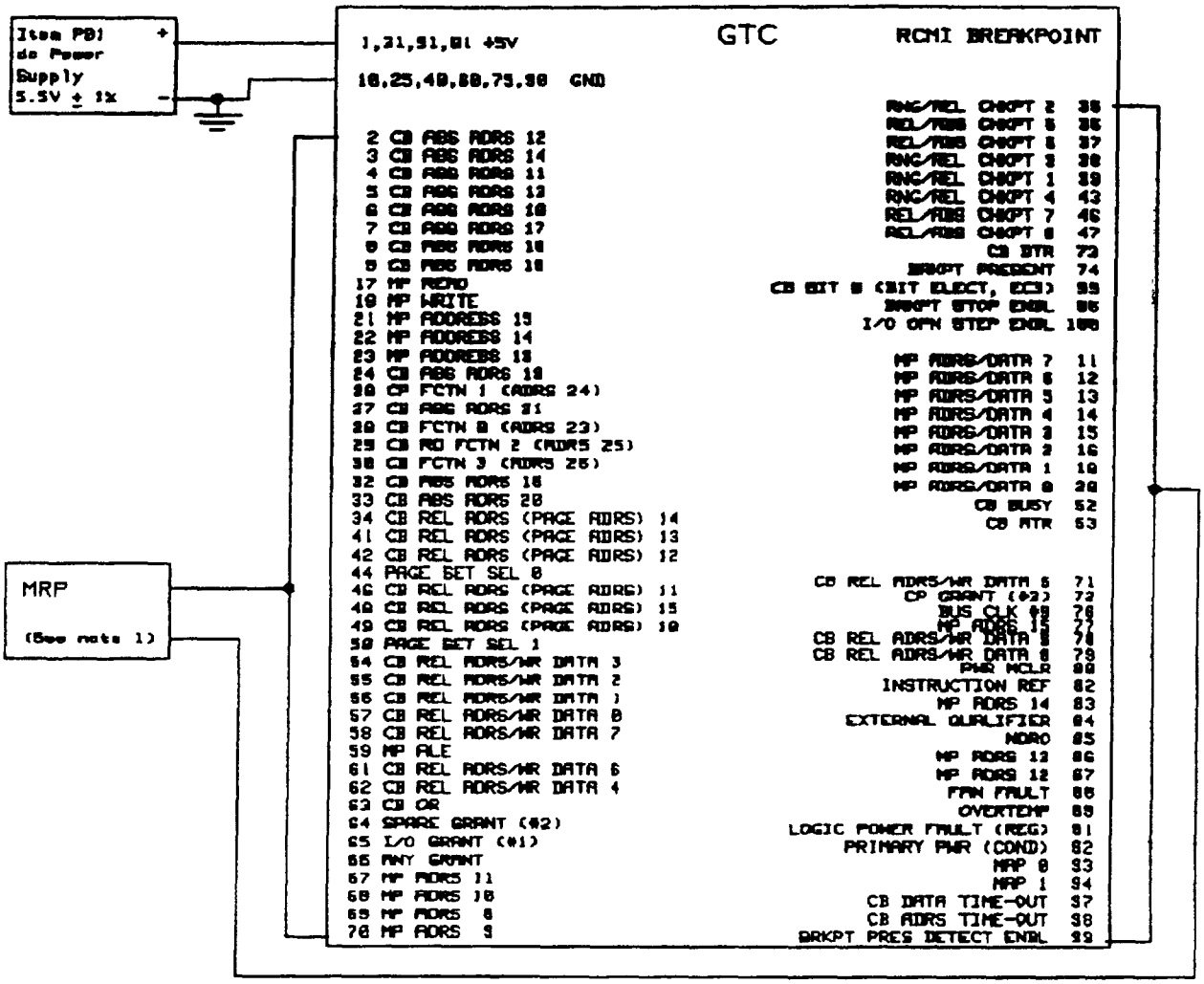
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. GTC life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 Vdc to 5.5 Vdc. Module power dissipation: 8.982 W (max); 6.021 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CB ABS A12	-1.2	0.4	2.4	-0.005	0.005	0.005	50
CB ABS A14	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB ABS A11	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB ABS A13	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB ABS A10	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB ABS A17	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB ABS A16	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB ABS A15	-1.2	0.4	2.4	-0.005	0.005	0.005	75
MP READ	-1.5	0.4	2.4	-0.5	0.05	2.0	125
MP WRITE	NA	0.4	2.4	-0.76	0.085	3.01	125
MP ADRS 15	-1.5	0.4	2.4	-0.51	0.051	2.0	75
MP ADRS 14	-1.5	0.4	2.4	-0.5	0.05	2.0	75
MP ADRS 13	-1.5	0.4	2.4	-0.25	0.025	1.0	50
CB ABS A19	-1.2	0.4	2.4	-0.005	0.005	0.005	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CB FCTN 1	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB ABS A21	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB FCTN 0	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB RD FCTN 2	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB FCTN 3	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB ABS A18	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB ABS A20	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB REL A14	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB REL A13	-1.2	0.4	2.4	-0.005	0.005	0.005	100
CB REL A12	-1.2	0.4	2.4	-0.005	0.005	0.005	75
PAGE SEL 0	-1.2	0.4	2.4	-0.005	0.005	0.005	100
CB REL A11	-1.2	0.4	2.4	-0.005	0.005	0.005	100
CB REL A15	-1.2	0.4	2.4	-0.005	0.005	0.005	125
CB REL A10	-1.2	0.4	2.4	-0.005	0.005	0.005	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
PAGE SEL 1	-1.2	0.4	2.4	-0.005	0.005	0.005	100
CB A/WR D3	-1.2	0.4	2.4	-0.005	0.005	0.005	50
CB A/WR D2	-1.2	0.4	2.4	-0.005	0.005	0.005	50
CB A/WR D1	-1.2	0.4	2.4	-0.005	0.005	0.005	50
CB A/WR D0	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB A/WR D7	-1.2	0.4	2.4	-0.005	0.005	0.005	75
MP ALE	NA	0.4	2.4	-0.63	0.05	0.13	175
CB A/WR D6	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB A/WR D4	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CB OR	-1.2	0.4	2.4	-0.005	0.005	0.005	75
SPARE GRANT	-1.5	0.4	2.4	-0.25	0.025	1.0	150
I/O GRANT	-1.5	0.4	2.4	-0.25	0.025	1.0	100
ANY GRANT	-1.5	0.4	2.4	-0.255	0.03	1.0	150
MP ADRS 11	NA	0.4	2.4	-0.265	0.04	1.01	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MP ADRS 10	NA	0.4	2.4	-0.27	0.045	1.02	125
MP ADRS 8	NA	0.4	2.4	-0.275	0.05	1.02	150
MP ADRS 9	NA	0.4	2.4	-0.27	0.045	1.02	125
CB A/WR D5	-1.2	0.4	2.4	-0.005	0.005	0.005	75
CP GRANT	-1.5	0.4	2.4	-0.25	0.025	1.0	125
BUS CLK	-1.2	0.4	2.4	-0.605	0.025	0.105	75
MP ADRS 15	-1.5	0.4	2.4	-0.5	0.05	2.0	75
CB A/WR D9	-1.2	0.4	2.4	-0.005	0.005	0.005	50
CB A/WR D8	-1.2	0.4	2.4	-0.005	0.005	0.005	50
PWR MCLR	NA	0.4	2.4	-2.86	0.1	1.31	217
MP ADRS 14	-1.5	0.4	2.4	-0.5	0.05	2.0	75
EX QUALIFIER	-1.2	0.4	2.4	-0.005	0.005	0.005	75
NDRO	-1.2	0.4	2.4	-0.005	0.005	0.005	75
MP ADRS 13	-1.5	0.4	2.4	-0.5	0.05	2.0	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MP ADRS 12	-1.5	0.4	2.4	-0.25	0.025	1.0	75
FAN FAULT	-1.5	0.4	2.4	-0.25	0.025	1.0	75
OVERTEMP	-1.5	0.4	2.4	-0.25	0.025	1.0	75
LOGIC PFAULT	-1.5	0.4	2.4	-0.25	0.025	1.0	75
PRIMARY PWR	-1.5	0.4	2.4	-0.25	0.025	1.0	75
MAP 0	-1.5	0.4	2.4	-0.25	0.025	1.0	75
MAP 1	-1.5	0.4	2.4	-0.25	0.025	1.0	75
CB DATA T/O	-1.5	0.4	2.4	-0.5	0.05	2.0	75
CB ADRS T/O	-1.5	0.4	2.4	-0.5	0.05	2.0	75
BP DETECT EN	-1.2	0.4	2.4	-2.0	0.05	0.1	75
MP A/D 7	NA	0.4	2.4	-0.2	0.02	NA	75
MP A/D 6	NA	0.4	2.4	-0.2	0.02	NA	75
MP A/D 5	NA	0.4	2.4	-0.2	0.02	NA	75
MP A/D 4	NA	0.4	2.4	-0.2	0.02	NA	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MP A/D 3	NA	0.4	2.4	-0.2	0.02	NA	75
MP A/D 2	NA	0.4	2.4	-0.2	0.02	NA	75
MP A/D 1	NA	0.4	2.4	-0.2	0.02	NA	75
MP A/D 0	NA	0.4	2.4	-0.2	0.02	NA	75
CB BUSY	NA	0.4	2.4	-0.015	0.015	NA	75
CB ATR	NA	0.4	2.4	-0.515	0.065	NA	200
INSTRTN REF	NA	0.4	2.4	-10.4	-5.52	0.23	100

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
MP A/D 7	2.4	0.5	-0.4	6.0	NA	NA	-40.0	75
MP A/D 6	2.4	0.5	-0.4	6.0	NA	NA	-40.0	75
MP A/D 5	2.4	0.5	-0.4	6.0	NA	NA	-40.0	75
MP A/D 4	2.4	0.5	-0.4	6.0	NA	NA	-40.0	75
MP A/D 3	2.4	0.5	-0.4	6.0	NA	NA	-40.0	75
MP A/D 2	2.4	0.5	-0.4	6.0	NA	NA	-40.0	75
MP A/D 1	2.4	0.5	-0.4	6.0	NA	NA	-40.0	75
MP A/D 0	2.4	0.5	-0.4	6.0	NA	NA	-40.0	75
CB BUSY	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
CB ATR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	200
RNG/REL CP 2	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
REL/ABS CP 6	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
REL/ABS CP 5	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
RNG/REL CP 3	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
RNG/REL CP 1	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
RNG/REL CP 4	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
REL/ABS CP 7	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
REL/ABS CP 8	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
CB DTR	2.4	0.5	-0.4	6.0	0.01	0.01	-60.0	NA
BRKPT PRESNT	2.4	0.5	NA	1.6	0.25	NA	NA	NA
CB BIT 0	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
BRKPT STOPEN	2.4	0.5	NA	1.6	0.5	NA	NA	NA
I/O OPN EN	2.4	0.5	NA	1.6	0.25	NA	NA	NA

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	BUS CLK to CB BUSY	-	-	-	45	45	45	nS	11
2	tpd	BUS CLK to CB ATR	-	-	-	45	45	45	nS	11
3	tpd	BUS CLK to CB DTR	-	-	-	46	46	46	nS	11
4	tpd	BP DETECT EN to BRKPT PRESNT	-	-	-	20	20	20	nS	11
5	tpd	MP A/D 0-3, 5-7 to BRKPT STOPEN	-	-	-	93	93	93	nS	11
6	tpd	MP A/D 4 to RNG/REL CP 1,3,4	-	-	-	101	101	101	nS	11
6	tpd	MP A/D 4 to REL/ABS CP 5-8	-	-	-	101	101	101	nS	11
7	tpd	MP READ to MP A/D 0-7	-	-	-	92	92	92	nS	11
8	tpd	MP A/D 5 to RNG/REL CP 2	-	-	-	101	101	101	nS	11

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6.4 Module function. The breakpoint module contains breakpoints and checkpoints, user-definable flags and hardware status discretes. Random Access Memories (RAMs) contained on the module are programmed by the Maintenance Processor (MP) to act as comparators. The maintenance processor stores ones at those addresses where the comparison is defined as a match and zeros elsewhere. The RAM data output gives the result of the address comparison.

6.4.1 Breakpoint register. A breakpoint register compares the state of the common bus function, grant, instruction reference and address lines against prestored selections. A match causes either a breakpoint or checkpoint. A checkpoint is defined as a breakpoint match that does not stop the Central Processor (CP) or Input/Output Controller (IOC), but flags the maintenance processor that an address was accessed. A breakpoint is a match that causes the CP or IOC(s) to stop. In either case, a First-In/First-Out (FIFO) memory is used to collect consecutive checkpoint and breakpoint addresses since the speed of the maintenance processor is much slower than the CP and IOC.

6.4.2 Maintenance processor. The MP RAM interface is used by the maintenance processor to program the static RAM address comparators with breakpoint addresses and functions. The common bus interface controls the path used to compare common bus addresses and functions by addressing the preprogrammed locations in the static RAM address comparators. The master write logic allows the breakpoint module to be slave on the common bus to four master write instructions from the CP. The MP bus interface is used by the maintenance processor to read/write all control registers, the scratch pad RAM and FIFO memories. The static RAM address comparators are the first stage of address comparison and are programmed by the maintenance processor with common bus addresses. These addresses are then used as address lines to the RAMs. A read of the preprogrammed addresses puts data on the RAM data lines which are further compared in the RAM data comparators to generate a checkpoint match.

6.4.3 FIFO memory. The FIFO memories are used to latch and hold up to 1024 common bus relative addresses and functions. The FIFOs are written into any time a checkpoint, breakpoint, or master write to the breakpoint module occurs. The scratchpad RAM is used by the maintenance processor as general purpose storage. The control logic is all logic that controls functions on the breakpoint module. The discrete buffers input the hardware descretes which flag abnormal conditions in the Data Processing System (DPS) hardware.

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6.4.4 Checkpoint enable generator. The checkpoint enable generator is part of the second stage of address comparison. It is used to qualify a checkpoint which, in turn, causes the CP or IOC to stop. The RAM data and checkpoint comparators use the data lines from the static RAM address comparators and checkpoint enables to perform the second stage of address comparison. This stage is the comparison summation of ranging, relative and absolute addresses. The breakpoint enable generator is part of the third state of address comparison and qualification of checkpoint comparison. The breakpoint comparators use the checkpoint compare result and the breakpoint enables to provide the CP or IOC stop signals.

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6.4.5 Description of I/O pins.

Signal name	Pin number	I/O	Description
CB ABS A12	2	I	CB ABS ADRS 12
CB ABS A14	3	I	CB ABS ADRS 14
CB ABS A11	4	I	CB ABS ADRS 11
CB ABS A13	5	I	CB ABS ADRS 13
CB ABS A10	6	I	CB ABS ADRS 10
CB ABS A17	7	I	CB ABS ADRS 17
CB ABS A16	8	I	CB ABS ADRS 16
CB ABS A15	9	I	CB ABS ADRS 15
MP READ	17	I	MP READ
MP WRITE	19	I	MP WRITE
MP ADRS 15	21	I	MP ADDRESS 15
MP ADRS 14	22	I	MP ADDRESS 14
MP ADRS 13	23	I	MP ADDRESS 13
CB ABS A19	24	I	CB ABS ADRS 19
CB FCTN 1	26	I	CB FCTN 1 (ADRS 24)
CB ABS A21	27	I	CB ABS ADRS 21
CB FCTN 0	28	I	CB FCTN 0 (ADRS 23)
CB RD FCTN 2	29	I	CB RD FCTN 2 (ADRS 25)
CB FCTN 3	30	I	CB FCTN 3 (ADRS 26)
CB ABS A18	32	I	CB ABS ADRS 18

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CB ABS A20	33	I	CB ABS ADRS 20
CB REL A14	34	I	CB REL ADRS (PAGE ADRS) 14
CB REL A13	41	I	CB REL ADRS (PAGE ADRS) 13
CB REL A12	42	I	CB REL ADRS (PAGE ADRS) 12
PAGE SEL 0	44	I	PAGE SET SEL 0
CB REL A11	46	I	CB REL ADRS (PAGE ADRS) 11
CB REL A15	48	I	CB REL ADRS (PAGE ADRS) 15
CB REL A10	49	I	CB REL ADRS (PAGE ADRS) 10
PAGE SEL 1	50	I	PAGE SET SEL 1
CB A/WR D3	54	I	CB REL ADRS/WR DATA 3
CB A/WR D2	55	I	CB REL ADRS/WR DATA 2
CB A/WR D1	56	I	CB REL ADRS/WR DATA 1
CB A/WR D0	57	I	CB REL ADRS/WR DATA 0
CB A/WR D7	58	I	CB REL ADRS/WR DATA 7
MP ALE	59	I	MP ALE
CB A/WR D6	61	I	CB REL ADRS/WR DATA 6
CB A/WR D4	62	I	CB REL ADRS/WR DATA 4
CB OR	63	I	CB OR
SPARE GRANT	64	I	SPARE GRANT (#2)
I/O GRANT	65	I	I/O GRANT (#1)

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
ANY GRANT	66	I	ANY GRANT
MP ADRS 11	67	I	MP ADRS 11
MP ADRS 10	68	I	MP ADRS 10
MP ADRS 8	69	I	MP ADRS 8
MP ADRS 9	70	I	MP ADRS 9
CB A/WR D5	71	I	CB REL ADRS/WR DATA 5
CP GRANT	72	I	CP GRANT (#3)
BUS CLK	76	I	BUS CLK #9
MP ADRS 15	77	I	MP ADRS 15
CB A/WR D9	78	I	CB REL ADRS/WR DATA 9
CB A/WR D8	79	I	CB REL ADRS/WR DATA 8
PWR MCLR	80	I	PWR MCLR
INST RTN REF	82	I	INSTRUCTION REF
MP ADRS 14	83	I	MP ADRS 14
EX QUALIFIER	84	I	EXTERNAL QUALIFIER
NDRO	85	I	NDRO
MP ADRS 13	86	I	MP ADRS 13
MP ADRS 12	87	I	MP ADRS 12
FAN FAULT	88	I	FAN FAULT
OVERTEMP	89	I	OVERTEMP
LOGIC PFAULT	91	I	LOGIC POWER FAULT (REG)

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
PRIMARY PWR	92	I	PRIMARY PWR (COND)
MAP 0	93	I	MAP 0
MAP 1	94	I	MAP 1
CB DATA T/O	97	I	CB DATA TIME-OUT
CB ADRS T/O	98	I	CB ADRS TIME-OUT
BP DETECT EN	99	I	BRKPT PRES DETECT ENBL
MP A/D 7	11	I/O	MP ADRS/DATA 7
MP A/D 6	12	I/O	MP ADRS/DATA 6
MP A/D 5	13	I/O	MP ADRS/DATA 5
MP A/D 4	14	I/O	MP ADRS/DATA 4
MP A/D 3	15	I/O	MP ADRS/DATA 3
MP A/D 2	16	I/O	MP ADRS/DATA 2
MP A/D 1	18	I/O	MP ADRS/DATA 1
MP A/D 0	20	I/O	MP ADRS/DATA 0
CB BUSY	52	I/O	CB BUSY
CB ATR	53	I/O	CB ATR
RNG/REL CP 2	35	O	RNG/REL CHKPT 2
REL/ABS CP 6	36	O	REL/ABS CHKPT 6
REL/ABS CP 5	37	O	REL/ABS CHKPT 5

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6.4.5 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
RNG/REL CP 3	38	O	RNG/REL CHKPT 3
RNG/REL CP 1	39	O	RNG/REL CHKPT 1
RNG/REL CP 4	43	O	RNG/REL CHKPT 4
REL/ABS CP 7	45	O	REL/ABS CHKPT 7
REL/ABS CP 8	47	O	REL/ABS CHKPT 8
CB DTR	73	O	CB DTR
BRKPT PRESNT	74	O	BRKPT PRESENT
CB BIT 0	95	O	CB BIT 0 (BIT ELECT, EC 3)
BRKPT STOPEN	96	O	BRKPT STOP ENBL
I/O OPN EN	100	O	I/O OPN STEP ENBL

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0263)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation	C	Frame	10	INF	MOhm	1-8

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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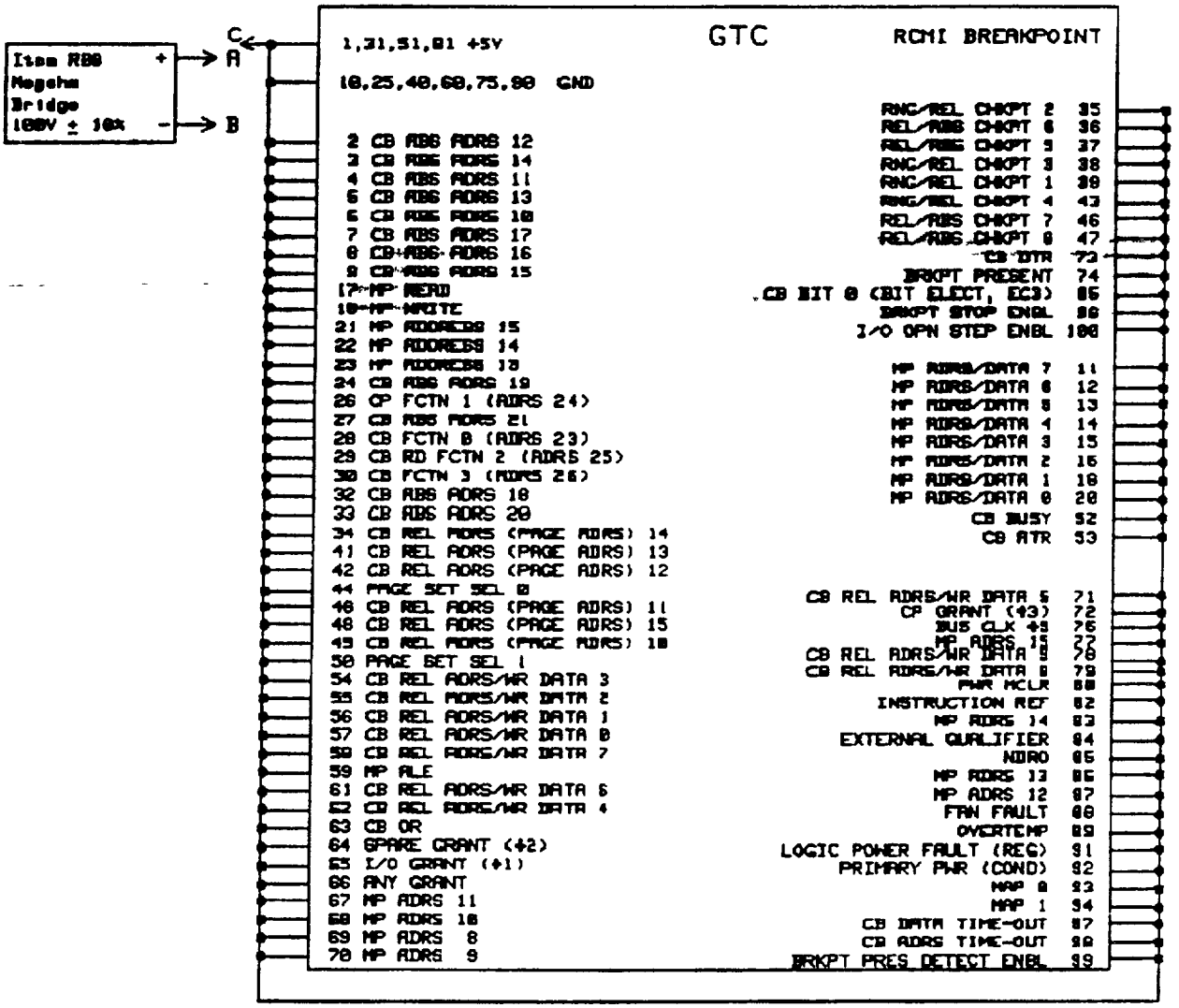


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	40	60				
B4	Continuity, GND	60	75				
B5	Continuity, GND	60	90				
B6	Continuity, Vcc	1	31				
B7	Continuity, Vcc	31	51				
B8	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B8

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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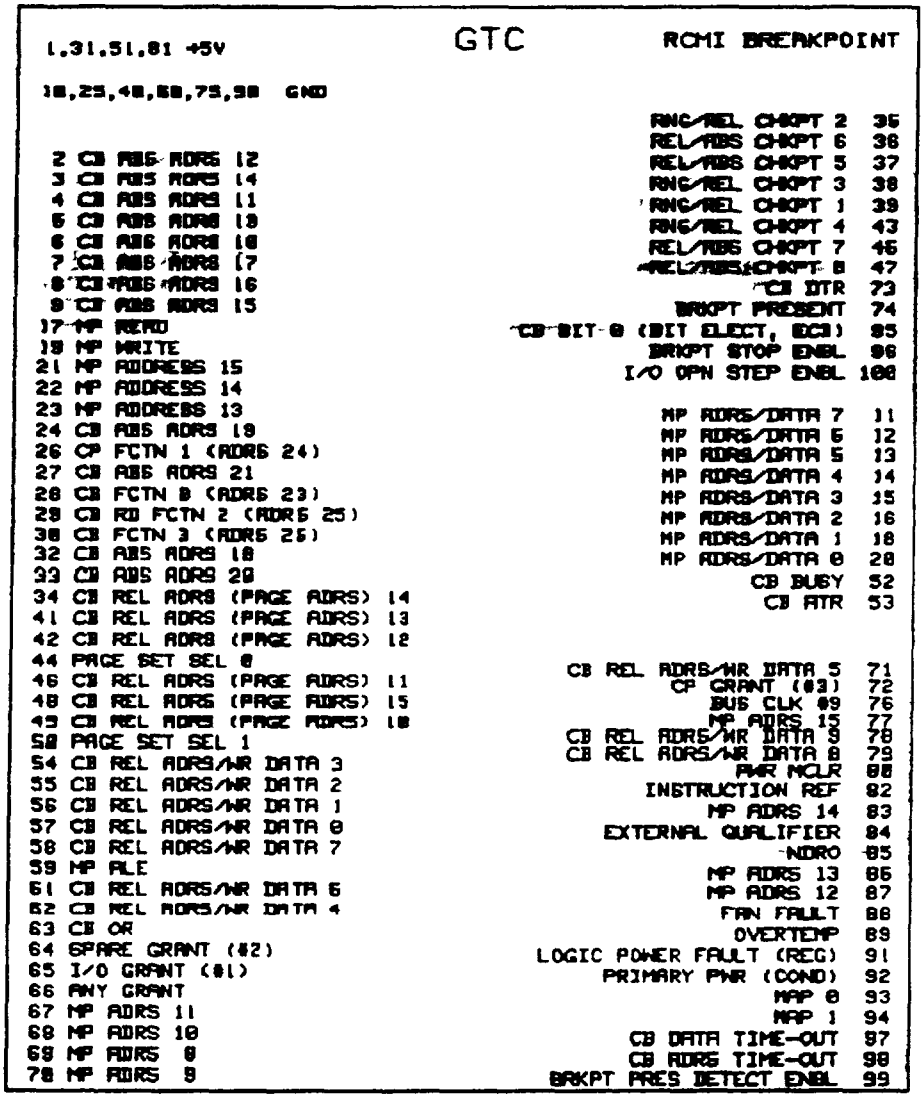


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.939A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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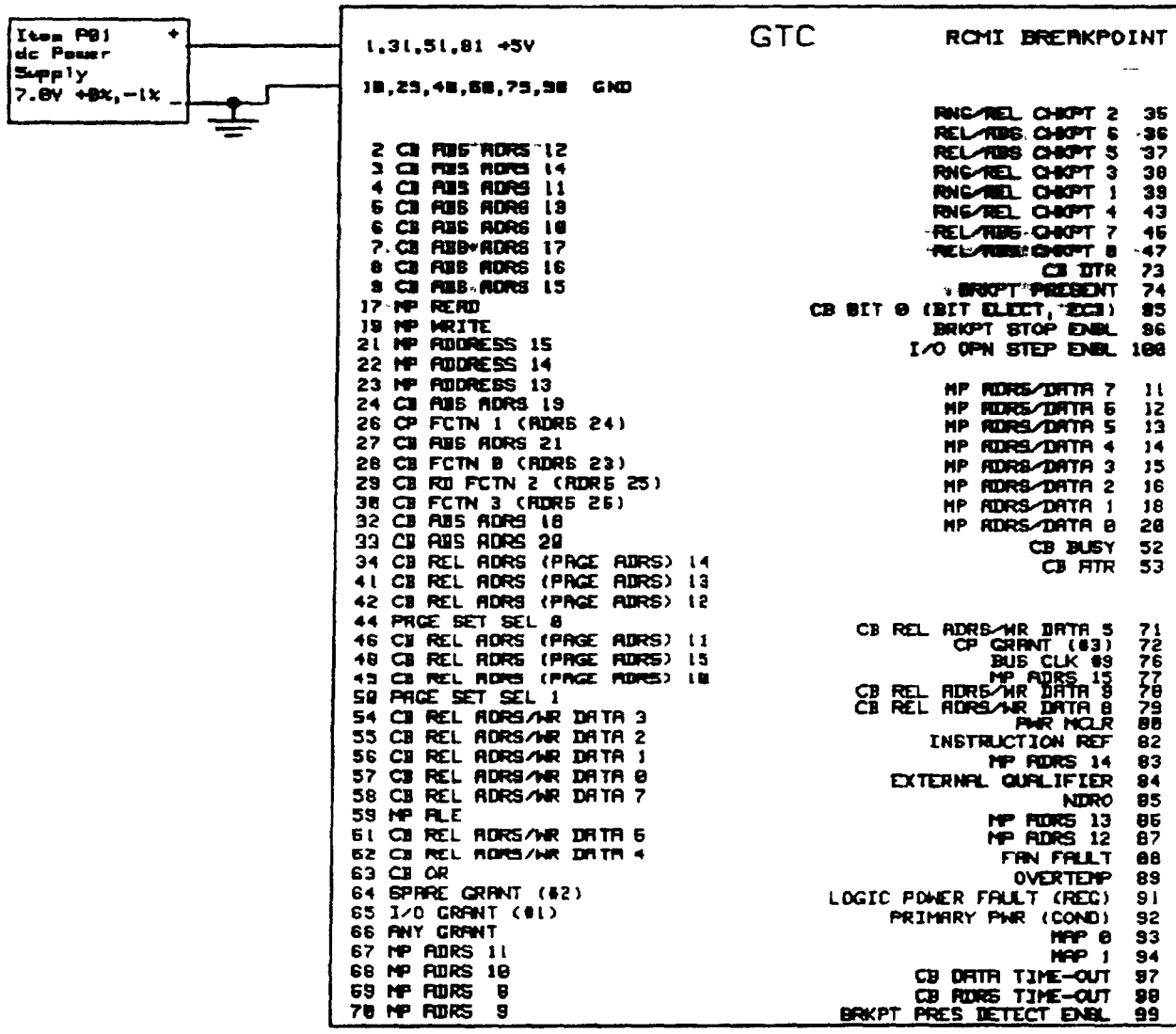


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	CB ABS A12	2	5.5	0.0	0.005	mA	1-3
D2	CB ABS A14	3					
D3	CB ABS A11	4					
D4	CB ABS A13	5					
D5	CB ABS A10	6					
D6	CB ABS A17	7					
D7	CB ABS A16	8					
D8	CB ABS A15	9	5.5	0.0	0.005	mA	1-3
D9	MP READ	17	5.5	0.0	2.0	mA	1-3
D10	MP WRITE	19	5.5	0.0	3.01	mA	1-3
D11	MP ADRS 15	21	5.5	0.0	2.0	mA	1-3
D12	MP ADRS 14	22	5.5	0.0	2.0	mA	1-3
D13	MP ADRS 13	23	5.5	0.0	1.0	mA	1-3
D14	CB ABS A19	24	5.5	0.0	0.005	mA	1-3
D15	CB FCTN 1	26					
D16	CB ABS A21	27					
D17	CB FCTN 0	28					
D18	CB RD FCTN 2	29					
D19	CB FCTN 3	30					
D20	CB ABS A18	32	5.5	0.0	0.005	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	CB ABS A20	33	5.5	0.0	0.005	mA	1-3
D22	CB REL A14	34					
D23	CB REL A13	41					
D24	CB REL A12	42					
D25	PAGE SEL 0	44					
D26	CB REL A11	46					
D27	CB REL A15	48					
D28	CB REL A10	49					
D29	PAGE SEL 1	50					
D30	CB A/WR D3	54					
D31	CB A/WR D2	55					
D32	CB A/WR D1	56					
D33	CB A/WR D0	57					
D34	CB A/WR D7	58	5.5	0.0	0.005	mA	1-3
D35	MP ALE	59	5.5	0.0	0.13	mA	1-3
D36	CB A/WR D6	61	5.5	0.0	0.005	mA	1-3
D37	CB A/WR D4	62					
D38	CB OR	63	5.5	0.0	0.005	mA	1-3
D39	SPARE GRANT	64	5.5	0.0	1.0	mA	1-3
D40	I/O GRANT	65	5.5	0.0	1.0	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	ANY GRANT	66	5.5	0.0	1.0	mA	1-3
D42	MP ADRS 11	67	5.5	0.0	1.01	mA	1-3
D43	MP ADRS 10	68	5.5	0.0	1.02	mA	1-3
D44	MP ADRS 8	69	|	|	|	|	|
D45	MP ADRS 9	70	5.5	0.0	1.02	mA	1-3
D46	CB A/WR D5	71	5.5	0.0	0.005	mA	1-3
D47	CP GRANT	72	5.5	0.0	1.0	mA	1-3
D48	BUS CLK	76	5.5	0.0	0.105	mA	1-3
D49	MP ADRS 15	77	5.5	0.0	2.0	mA	1-3
D50	CB A/WR D9	78	5.5	0.0	0.005	mA	1-3
D51	CB A/WR D8	79	5.5	0.0	0.005	mA	1-3
D52	PWR MCLR	80	5.5	0.0	1.31	mA	1-3
D53	INSTRTN REF	82	5.5	-0.22	0.23	mA	1-3
D54	MP ADRS 14	83	5.5	0.0	2.0	mA	1-3
D55	EX QUALIFIER	84	5.5	0.0	0.005	mA	1-3
D56	NDRO	85	5.5	0.0	0.005	mA	1-3
D57	MP ADRS 13	86	5.5	0.0	2.0	mA	1-3
D58	MP ADRS 12	87	5.5	0.0	1.0	mA	1-3
D59	FAN FAULT	88	|	|	|	|	|
D60	OVERTEMP	89	5.5	0.0	1.0	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D61	LOGIC PFAULT	91	5.5	0.0	1.0	mA	1-3
D62	PRIMARY PWR	92					
D63	MAP 0	93					
D64	MAP 1	94	5.5	0.0	1.0	mA	1-3
D65	CB DATA T/O	97	5.5	0.0	2.0	mA	1-3
D66	CB ADRS T/O	98	5.5	0.0	2.0	mA	1-3
D67	BP DETECT EN	99	5.5	0.0	0.1	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D67
Power supply current limit	2.45 A	NA	D1-D67
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D67
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D67 D1-D67
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 0.25 uA ± 100.0 uA ± 150.0 uA ± 100.0 uA ± 50.0 uA ± 0.25 uA ± 6.5 uA ± 0.25 uA ± 50.0 uA ± 50.5 uA ± 51.0 uA ± 0.25 uA ± 50.0 uA ± 5.25 uA ± 100.0 uA ± 0.25 uA ± 65.5 uA ± 22.5 uA ± 100.0 uA ± 0.25 uA ± 100.0 uA ± 50.0 uA ± 100.0 uA ± 5.0 uA	D1-D8 D9 D10 D11,D12 D13 D14-D34 D35 D36-D38 D39-D41 D42 D43-D45 D46 D47 D48 D49 D50,D51 D52 D53 D54 D55,D56 D57 D58-D64 D65,D66 D67

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	MP A/D 7	11	0.0	-240.0	-40.0	mA	1-4
E2	MP A/D 6	12					
E3	MP A/D 5	13					
E4	MP A/D 4	14					
E5	MP A/D 3	15					
E6	MP A/D 2	16					
E7	MP A/D 1	18					
E8	MP A/D 0	20	0.0	-240.0	-40.0	mA	1-4
E9	RNG/REL CP 2	35	0.0	-250.0	-30.0	mA	1-4
E10	REL/ABS CP 6	36					
E11	REL/ABS CP 5	37					
E12	RNG/REL CP 3	38					
E13	RNG/REL CP 1	39					
E14	RNG/REL CP 4	43					
E15	REL/ABS CP 7	45					
E16	REL/ABS CP 8	47	0.0	-250.0	-30.0	mA	1-4
E17	CB BUSY	52	0.0	-240.0	-60.0	mA	1-4
E18	CB ATR	53					
E19	CB DTR	73	0.0	-240.0	-60.0	mA	1-4
E20	CB BIT 0	95	0.0	-250.0	-30.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E20
Power supply current limit	2.45 A	NA	E1-E20
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E20
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E20 E1-E20
Output current measurement	NA NA NA NA	± 10 mA ± 11 mA ± 9 mA ± 11 mA	E1-E8 E9-E16 E17-E19 E20

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	CB ABS A12	2	2.4	-0.005	0.005	mA	1-3
F2	CB ABS A14	3					
F3	CB ABS A11	4					
F4	CB ABS A13	5					
F5	CB ABS A10	6					
F6	CB ABS A17	7					
F7	CB ABS A16	8					
F8	CB ABS A15	9	2.4	-0.005	0.005	mA	1-3
F9	MP A/D 7	11	2.4	-0.2	0.02	mA	1-3
F10	MP A/D 6	12					
F11	MP A/D 5	13					
F12	MP A/D 4	14					
F13	MP A/D 3	15					
F14	MP A/D 2	16	2.4	-0.2	0.02	mA	1-3
F15	MP READ	17	2.4	-0.5	0.05	mA	1-3
F16	MP A/D 1	18	2.4	-0.2	0.02	mA	1-3
F17	MP WRITE	19	2.4	-0.76	0.085	mA	1-3
F18	MP A/D 0	20	2.4	-0.2	0.02	mA	1-3
F19	MP ADRS 15	21	2.4	-0.51	0.051	mA	1-3
F20	MP ADRS 14	22	2.4	-0.5	0.05	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	MP ADRS 13	23	2.4	-0.25	0.025	mA	1-3
F22	CB ABS A19	24	2.4	-0.005	0.005	mA	1-3
F23	CB FCTN 1	26					
F24	CB ABS A21	27					
F25	CB FCTN 0	28					
F26	CB RD FCTN 2	29					
F27	CB FCTN 3	30					
F28	CB ABS A18	32					
F29	CB ABS A20	33					
F30	CB REL A14	34					
F31	CB REL A13	41					
F32	CB REL A12	42					
F33	PAGE SEL 0	44					
F34	CB REL A11	46					
F35	CB REL A15	48					
F36	CB REL A10	49					
F37	PAGE SEL 1	50	2.4	-0.005	0.005	mA	1-3
F38	CB BUSY	52	2.4	-0.015	0.015	mA	1-3
F39	CB ATR	53	2.4	-0.515	0.065	mA	1-3
F40	CB A/WR D3	54	2.4	-0.005	0.005	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	CB A/WR D2	55	2.4	-0.005	0.005	mA	1-3
F42	CB A/WR D1	56					
F43	CB A/WR D0	57					
F44	CB A/WR D7	58	2.4	-0.005	0.005	mA	1-3
F45	MP ALE	59	2.4	-0.63	0.05	mA	1-3
F46	CB A/WR D6	61	2.4	-0.005	0.005	mA	1-3
F47	CB A/WR D4	62					
F48	CB OR	63	2.4	-0.005	0.005	mA	1-3
F49	SPARE GRANT	64	2.4	-0.25	0.025	mA	1-3
F50	I/O GRANT	65	2.4	-0.25	0.025	mA	1-3
F51	ANY GRANT	66	2.4	-0.255	0.03	mA	1-3
F52	MP ADRS 11	67	2.4	-0.265	0.04	mA	1-3
F53	MP ADRS 10	68	2.4	-0.27	0.045	mA	1-3
F54	MP ADRS 8	69	2.4	-0.275	0.05	mA	1-3
F55	MP ADRS 9	70	2.4	-0.27	0.045	mA	1-3
F56	CB A/WR D5	71	2.4	-0.005	0.005	mA	1-3
F57	CP GRANT	72	2.4	-0.25	0.025	mA	1-3
F58	BUS CLK	76	2.4	-0.605	0.025	mA	1-3
F59	MP ADRS 15	77	2.4	-0.5	0.05	mA	1-3
F60	CB A/WR D9	78	2.4	-0.005	0.005	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F61	CB A/WR D8	79	2.4	-0.005	0.005	mA	1-3
F62	PWR MCLR	80	2.4	-2.86	0.1	mA	1-3
F63	INSTRTN REF	82	2.4	-10.4	-5.52	mA	1-3
F64	MP ADRS 14	83	2.4	-0.5	0.05	mA	1-3
F65	EX QUALIFIER	84	2.4	-0.005	0.005	mA	1-3
F66	NDRO	85	2.4	-0.005	0.005	mA	1-3
F67	MP ADRS 13	86	2.4	-0.5	0.05	mA	1-3
F68	MP ADRS 12	87	2.4	-0.25	0.025	mA	1-3
F69	FAN FAULT	88					
F70	OVERTEMP	89					
F71	LOGIC PFAULT	91					
F72	PRIMARY PWR	92					
F73	MAP 0	93					
F74	MAP 1	94	2.4	-0.25	0.025	mA	1-3
F75	CB DATA T/O	97	2.4	-0.5	0.05	mA	1-3
F76	CB ADRS T/O	98	2.4	-0.5	0.05	mA	1-3
F77	BP DETECT EN	99	2.4	-2.0	0.05	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F77
Power supply current limit	2.45 A	NA	F1-F77
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F77
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F77 F1-F77
Input current measurement	NA	± 0.5 uA	F1-F8
	NA	± 11.0 uA	F9-F14
	NA	± 27.5 uA	F15
	NA	± 11.0 uA	F16
	NA	± 42.2 uA	F17
	NA	± 11.0 uA	F18
	NA	± 28.1 uA	F19
	NA	± 27.5 uA	F20
	NA	± 13.7 uA	F21
	NA	± 0.5 uA	F22-F37
	NA	± 1.5 uA	F38
	NA	± 29.0 uA	F39
	NA	± 0.5 uA	F40-F44
	NA	± 34.0 uA	F45
	NA	± 0.5 uA	F46-F48
	NA	± 13.7 uA	F49,F50
	NA	± 14.25 uA	F51
	NA	± 15.25 uA	F52
	NA	± 15.7 uA	F53
	NA	± 16.25 uA	F54
	NA	± 15.7 uA	F55
	NA	± 0.5 uA	F56
	NA	± 13.7 uA	F57
	NA	± 31.5 uA	F58

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
	NA	± 27.5 uA	F59
	NA	± 0.5 uA	F60,F61
	NA	± 148.0 uA	F62
	NA	± 244.0 uA	F63
	NA	± 27.5 uA	F64
	NA	± 0.5 uA	F65,F66
	NA	± 27.5 uA	F67
	NA	± 13.7 uA	F68-F74
	NA	± 27.5 uA	F75,F76
	NA	± 102.0 uA	F77

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	CB ABS A12	2	0.4	-0.005	0.005	mA	1-3
G2	CB ABS A14	3					
G3	CB ABS A11	4					
G4	CB ABS A13	5					
G5	CB ABS A10	6					
G6	CB ABS A17	7					
G7	CB ABS A16	8					
G8	CB ABS A15	9	0.4	-0.005	0.005	mA	1-3
G9	MP A/D 7	11	0.4	-0.2	0.02	mA	1-3
G10	MP A/D 6	12					
G11	MP A/D 5	13					
G12	MP A/D 4	14					
G13	MP A/D 3	15					
G14	MP A/D 2	16	0.4	-0.2	0.02	mA	1-3
G15	MP READ	17	0.4	-0.5	0.05	mA	1-3
G16	MP A/D 1	18	0.4	-0.2	0.02	mA	1-3
G17	MP WRITE	19	0.4	-0.76	0.085	mA	1-3
G18	MP A/D 0	20	0.4	-0.2	0.02	mA	1-3
G19	MP ADRS 15	21	0.4	-0.51	0.051	mA	1-3
G20	MP ADRS 14	22	0.4	-0.5	0.05	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	MP ADRS 13	23	0.4	-0.25	0.025	mA	1-3
G22	CB ABS A19	24	0.4	-0.005	0.005	mA	1-3
G23	CB FCTN 1	26					
G24	CB ABS A21	27					
G25	CB FCTN 0	28					
G26	CB RD FCTN 2	29					
G27	CB FCTN 3	30					
G28	CB ABS A18	32					
G29	CB ABS A20	33					
G30	CB REL A14	34					
G31	CB REL A13	41					
G32	CB REL A12	42					
G33	PAGE SEL 0	44					
G34	CB REL A11	46					
G35	CB REL A15	48					
G36	CB REL A10	49					
G37	PAGE SEL 1	50	0.4	-0.005	0.005	mA	1-3
G38	CB BUSY	52	0.4	-0.015	0.015	mA	1-3
G39	CB ATR	53	0.4	-0.515	0.065	mA	1-3
G40	CB A/WR D3	54	0.4	-0.005	0.005	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	CB A/WR D2	55	0.4	-0.005	0.005	mA	1-3
G42	CB A/WR D1	56					
G43	CB A/WR D0	57					
G44	CB A/WR D7	58	0.4	-0.005	0.005	mA	1-3
G45	MP ALE	59	0.4	-0.63	0.05	mA	1-3
G46	CB A/WR D6	61	0.4	-0.005	0.005	mA	1-3
G47	CB A/WR D4	62					
G48	CB OR	63	0.4	-0.005	0.005	mA	1-3
G49	SPARE GRANT	64	0.4	-0.25	0.025	mA	1-3
G50	I/O GRANT	65	0.4	-0.25	0.025	mA	1-3
G51	ANY GRANT	66	0.4	-0.255	0.03	mA	1-3
G52	MP ADRS 11	67	0.4	-0.265	0.04	mA	1-3
G53	MP ADRS 10	68	0.4	-0.27	0.045	mA	1-3
G54	MP ADRS 8	69	0.4	-0.275	0.05	mA	1-3
G55	MP ADRS 9	70	0.4	-0.27	0.045	mA	1-3
G56	CB A/WR D5	71	0.4	-0.005	0.005	mA	1-3
G57	CP GRANT	72	0.4	-0.25	0.025	mA	1-3
G58	BUS CLK	76	0.4	-0.605	0.025	mA	1-3
G59	MP ADRS 15	77	0.4	-0.5	0.05	mA	1-3
G60	CB A/WR D9	78	0.4	-0.005	0.005	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G61	CB A/WR D8	79	0.4	-0.005	0.005	mA	1-3
G62	PWR MCLR	80	0.4	-2.86	0.1	mA	1-3
G63	INSTRTN REF	82	0.4	-10.4	-5.52	mA	1-3
G64	MP ADRS 14	83	0.4	-0.5	0.05	mA	1-3
G65	EX QUALIFIER	84	0.4	-0.005	0.005	mA	1-3
G66	NDRO	85	0.4	-0.005	0.005	mA	1-3
G67	MP ADRS 13	86	0.4	-0.5	0.05	mA	1-3
G68	MP ADRS 12	87	0.4	-0.25	0.025	mA	1-3
G69	FAN FAULT	88					
G70	OVERTEMP	89					
G71	LOGIC PFAULT	91					
G72	PRIMARY PWR	92					
G73	MAP 0	93					
G74	MAP 1	94	0.4	-0.25	0.025	mA	1-3
G75	CB DATA T/O	97	0.4	-0.5	0.05	mA	1-3
G76	CB ADRS T/O	98	0.4	-0.5	0.05	mA	1-3
G77	BP DETECT EN	99	0.4	-2.0	0.05	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G77
Power supply current limit	2.45 A	NA	G1-G77
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G77
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G77
low, (VL)	0.0 V	± 100 mV	G1-G77
Input current measurement	NA	± 0.5 uA	G1-G8
	NA	± 11.0 uA	G9-G14
	NA	± 27.5 uA	G15
	NA	± 11.0 uA	G16
	NA	± 42.2 uA	G17
	NA	± 11.0 uA	G18
	NA	± 28.1 uA	G19
	NA	± 27.5 uA	G20
	NA	± 13.7 uA	G21
	NA	± 0.5 uA	G22-G37
	NA	± 1.5 uA	G38
	NA	± 29.0 uA	G39
	NA	± 0.5 uA	G40-G44
	NA	± 34.0 uA	G45
	NA	± 0.5 uA	G46-G48
	NA	± 13.7 uA	G49,G50
	NA	± 14.25 uA	G51
	NA	± 15.25 uA	G52
	NA	± 15.7 uA	G53
	NA	± 16.25 uA	G54
	NA	± 15.7 uA	G55
	NA	± 0.5 uA	G56
	NA	± 13.7 uA	G57
	NA	± 31.5 uA	G58

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
	NA	± 27.5 uA	G59
	NA	± 0.5 uA	G60,G61
	NA	± 148.0 uA	G62
	NA	± 244.0 uA	G63
	NA	± 27.5 uA	G64
	NA	± 0.5 uA	G65,G66
	NA	± 27.5 uA	G67
	NA	± 13.7 uA	G68-G74
	NA	± 27.5 uA	G75,G76
	NA	± 102.0 uA	G77

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	MP A/D 7	11	-0.4	2.4	4.5	V	1-4
H2	MP A/D 6	12					
H3	MP A/D 5	13					
H4	MP A/D 4	14					
H5	MP A/D 3	15					
H6	MP A/D 2	16					
H7	MP A/D 1	18					
H8	MP A/D 0	20	-0.4				
H9	RNG/REL CP 2	35	-0.16				
H10	REL/ABS CP 6	36					
H11	REL/ABS CP 5	37					
H12	RNG/REL CP 3	38					
H13	RNG/REL CP 1	39					
H14	RNG/REL CP 4	43					
H15	REL/ABS CP 7	45					
H16	REL/ABS CP 8	47	-0.16				
H17	CB BUSY	52	-0.4				
H18	CB ATR	53					
H19	CB DTR	73	-0.4				
H20	CB BIT 0	95	-0.16	2.4	4.5	V	1-4

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Check the source to verify that this is the current version before use.

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H20
Power supply current limit	2.45 A	NA	H1-H20
Forcing current, (Ioh)	-0.4 mA -0.16 mA -0.4 mA -0.16 mA	± 40 uA ± 16 uA ± 40 uA ± 16 uA	H1-H8 H9-H16 H17-H19 H20
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H20 H1-H20
Output voltage measurement	NA	± 105 mV	H1-H20

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

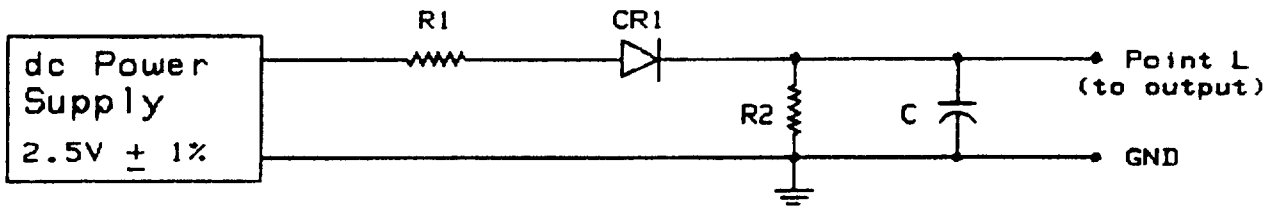
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
11,12,13,14,15,16, 18,20,52,53,73	6200	6.0	0.5
35,36,37,38,39,43, 45,47,95	16000	1.6	0.5
74,96,100	NA	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	MP A/D 7	11	6.0	0.0	0.5	V	1-4
J2	MP A/D 6	12					
J3	MP A/D 5	13					
J4	MP A/D 4	14					
J5	MP A/D 3	15					
J6	MP A/D 2	16					
J7	MP A/D 1	18					
J8	MP A/D 0	20	6.0				
J9	RNG/REL CP 2	35	1.6				
J10	REL/ABS CP 6	36					
J11	REL/ABS CP 5	37					
J12	RNG/REL CP 3	38					
J13	RNG/REL CP 1	39					
J14	RNG/REL CP 4	43					
J15	REL/ABS CP 7	45					
J16	REL/ABS CP 8	47	1.6				
J17	CB BUSY	52	6.0				
J18	CB ATR	53					
J19	CB DTR	73	6.0				
J20	BRKPT PRESNT	74	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	CB BIT 0	95	1.6	0.0	0.5	V	1-4
J22	BRKPT STOPEN	96					
J23	BRKPT STOPEN	96					
J24	I/O OPN EN	100	1.6	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:17Z
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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J24
Power supply current limit	2.45 A	NA	J1-J24
Forcing current, (Iol)	6.0 mA 1.6 mA 6.0 mA 1.6 mA	± 600 uA ± 160 uA ± 600 uA ± 160 uA	J1-J8 J9-J16 J17-J19 J20-J24
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J24 J1-J24
Output voltage measurement	NA	± 25 mV	J1-J24

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
High level							
K1	CB DTR	73	2.4	-0.01	0.01	mA	1-3
K2	BRKPT PRESNT	74	2.4	-0.25	0.25	mA	1-3
K3	BRKPT STOPEN	96	2.4	-0.5	0.5	mA	1-3
K4	I/O OPN EN	100	2.4	-0.25	0.25	mA	1-3
Low level							
K5	CB DTR	73	0.4	-0.01	0.01	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K5
Power supply current limit	2.45 A	NA	K1-K5
Forcing voltage, (Voh) (Vol)	2.4 V 0.4 V	± 50 mV ± 50 mV	K1-K4 K5
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	K1-K5 K1-K5
Output current measurement	NA NA NA NA NA	± 1 uA ± 25 uA ± 50 uA ± 25 uA ± 1 uA	K1 K2 K3 K4 K5

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	CB ABS A12	2	-18.0	-1.2	0.0	V	1-3
L2	CB ABS A14	3					
L3	CB ABS A11	4					
L4	CB ABS A13	5					
L5	CB ABS A10	6					
L6	CB ABS A17	7					
L7	CB ABS A16	8					
L8	CB ABS A15	9	-18.0	-1.2	0.0	V	1-3
L9	MP READ	17	-18.0	-1.5	0.0	V	1-3
L10	MP ADRS 15	21					
L11	MP ADRS 14	22					
L12	MP ADRS 13	23	-18.0	-1.5	0.0	V	1-3
L13	CB ABS A19	24	-18.0	-1.2	0.0	V	1-3
L14	CB FCTN 1	26					
L15	CB ABS A21	27					
L16	CB FCTN 0	28					
L17	CB RD FCTN 2	29					
L18	CB FCTN 3	30					
L19	CB ABS A18	32					
L20	CB ABS A20	33	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	CB REL A14	34	-18.0	-1.2	0.0	V	1-3
L22	CB REL A13	41					
L23	CB REL A12	42					
L24	PAGE SEL 0	44					
L25	CB REL A11	46					
L26	CB REL A15	48					
L27	CB REL A10	49					
L28	PAGE SEL 1	50					
L29	CB A/WR D3	54					
L30	CB A/WR D2	55					
L31	CB A/WR D1	56					
L32	CB A/WR D0	57					
L33	CB A/WR D7	58					
L34	CB A/WR D6	61					
L35	CB A/WR D4	62					
L36	CB OR	63	-18.0	-1.2	0.0	V	1-3
L37	SPARE GRANT	64	-18.0	-1.5	0.0	V	1-3
L38	I/O GRANT	65					
L39	ANY GRANT	66	-18.0	-1.5	0.0	V	1-3
L40	CB A/WR D5	71	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L41	CP GRANT	72	-18.0	-1.5	0.0	V	1-3
L42	BUS CLK	76	-18.0	-1.2	0.0	V	1-3
L43	MP ADRS 15	77	-18.0	-1.5	0.0	V	1-3
L44	CB A/WR D9	78	-18.0	-1.2	0.0	V	1-3
L45	CB A/WR D8	79	-18.0	-1.2	0.0	V	1-3
L46	MP ADRS 14	83	-18.0	-1.5	0.0	V	1-3
L47	EX QUALIFIER	84	-18.0	-1.2	0.0	V	1-3
L48	NDRO	85	-18.0	-1.2	0.0	V	1-3
L49	MP ADRS 13	86	-18.0	-1.5	0.0	V	1-3
L50	MP ADRS 12	87					
L51	FAN FAULT	88					
L52	OVERTEMP	89					
L53	LOGIC PFAULT	91					
L54	PRIMARY PWR	92					
L55	MAP 0	93					
L56	MAP 1	94					
L57	CB DATA T/O	97					
L58	CB ADRS T/O	98	-18.0	-1.5	0.0	V	1-3
L59	BP DETECT EN	99	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L59
Power supply current limit	NA	NA	L1-L59
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L59
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L59 L1-L59
Input voltage measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA	± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV	L1-L8 L9-L12 L13-L36 L37-L39 L40 L41 L42 L43 L44,L45 L46 L47,L48 L49-L58 L59

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M	PSFC	1	6.3	11.7	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M
Capacitance measurement	NA	± 270 nF	M

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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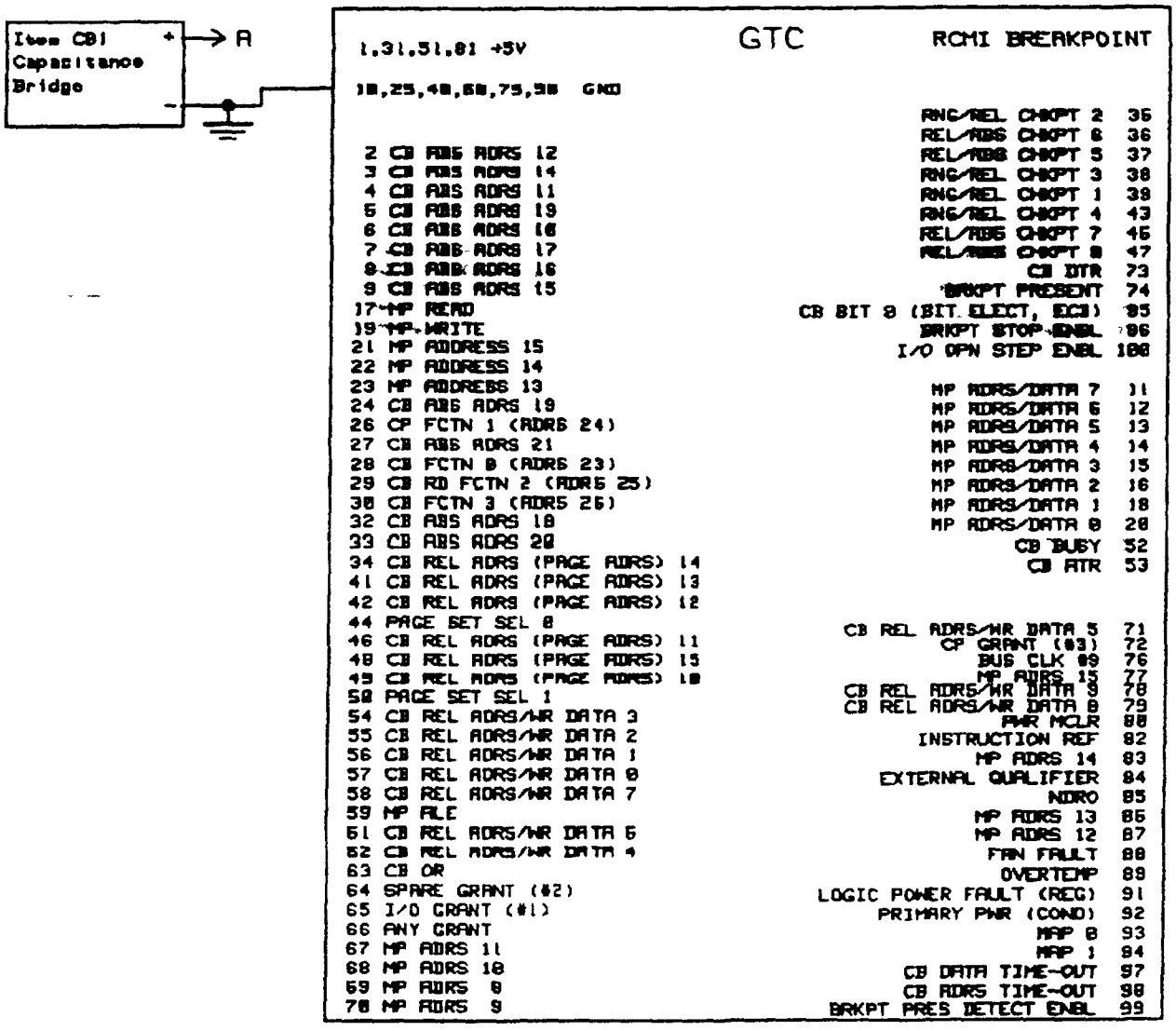


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31 51,81	680	1633	mA	1-3

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NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	2.45 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 47.6 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (325 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (325 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	2.45 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

3. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (325 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (325 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (325 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (325 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	2.45 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

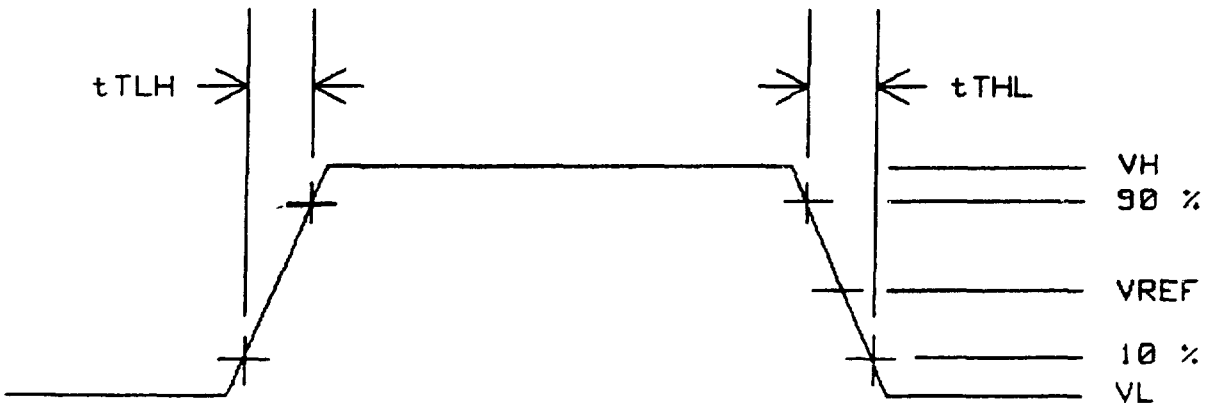
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 11. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	BUS CLK to CB BUSY	-	-	-	45	45	45	nS	11
2	tpd	BUS CLK to CB ATR	-	-	-	45	45	45	nS	11
3	tpd	BUS CLK to CB DTR	-	-	-	46	46	46	nS	11
4	tpd	BP DETECT EN to BRKPT PRESNT	-	-	-	20	20	20	nS	11
5	tpd	MP A/D 0-3, 5-7 to BRKPT STOPEN	-	-	-	93	93	93	nS	11
6	tpd	MP A/D 4 to RNG/REL CP 1,3,4	-	-	-	101	101	101	nS	11
6	tpd	MP A/D 4 to REL/ABS CP 5-8	-	-	-	101	101	101	nS	11
7	tpd	MP READ to MP A/D 0-7	-	-	-	92	92	92	nS	11
8	tpd	MP A/D 5 to RNG/REL CP 2	-	-	-	101	101	101	nS	11

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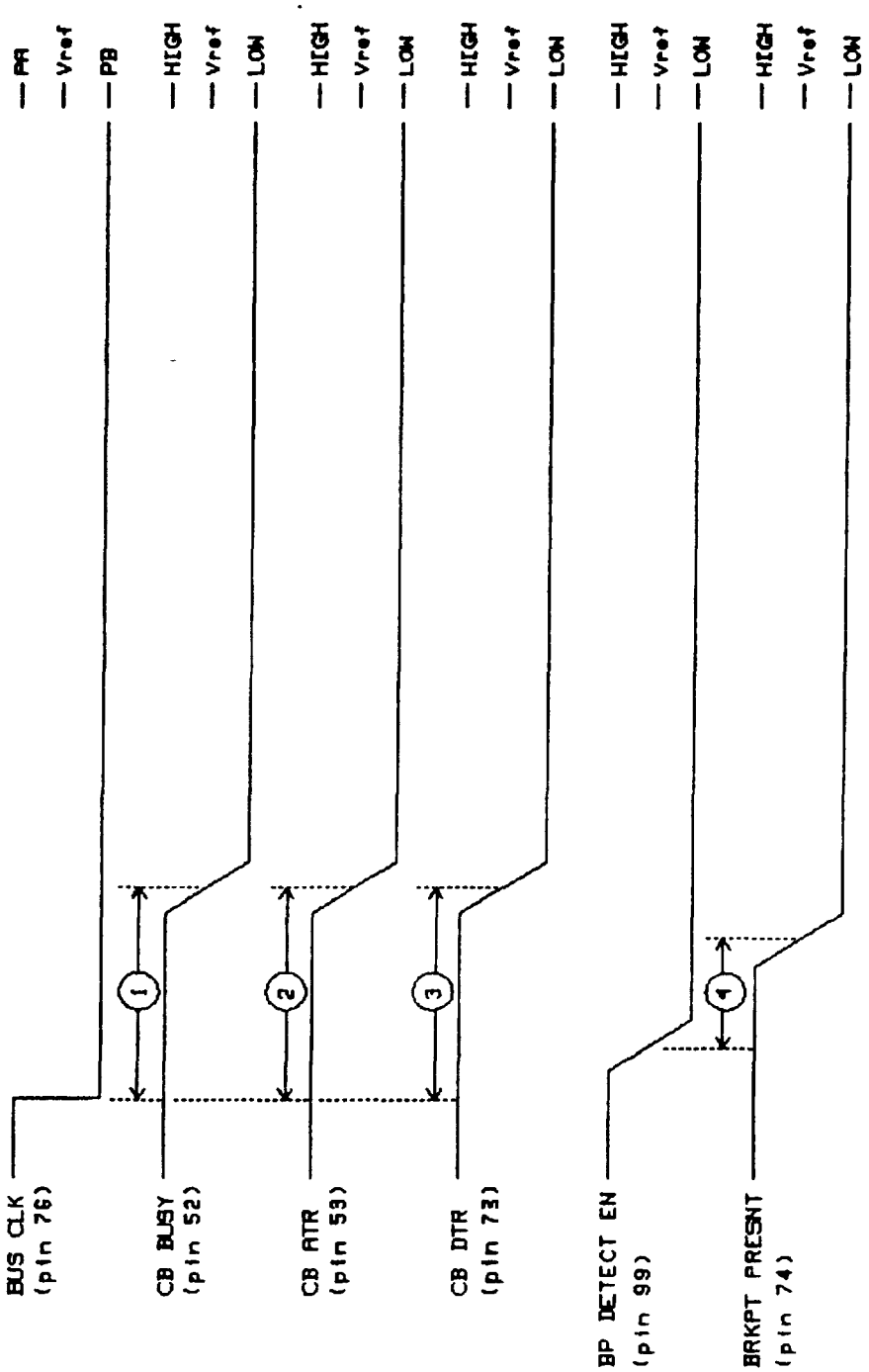


FIGURE 11a. Dynamic waveforms for table XVII.

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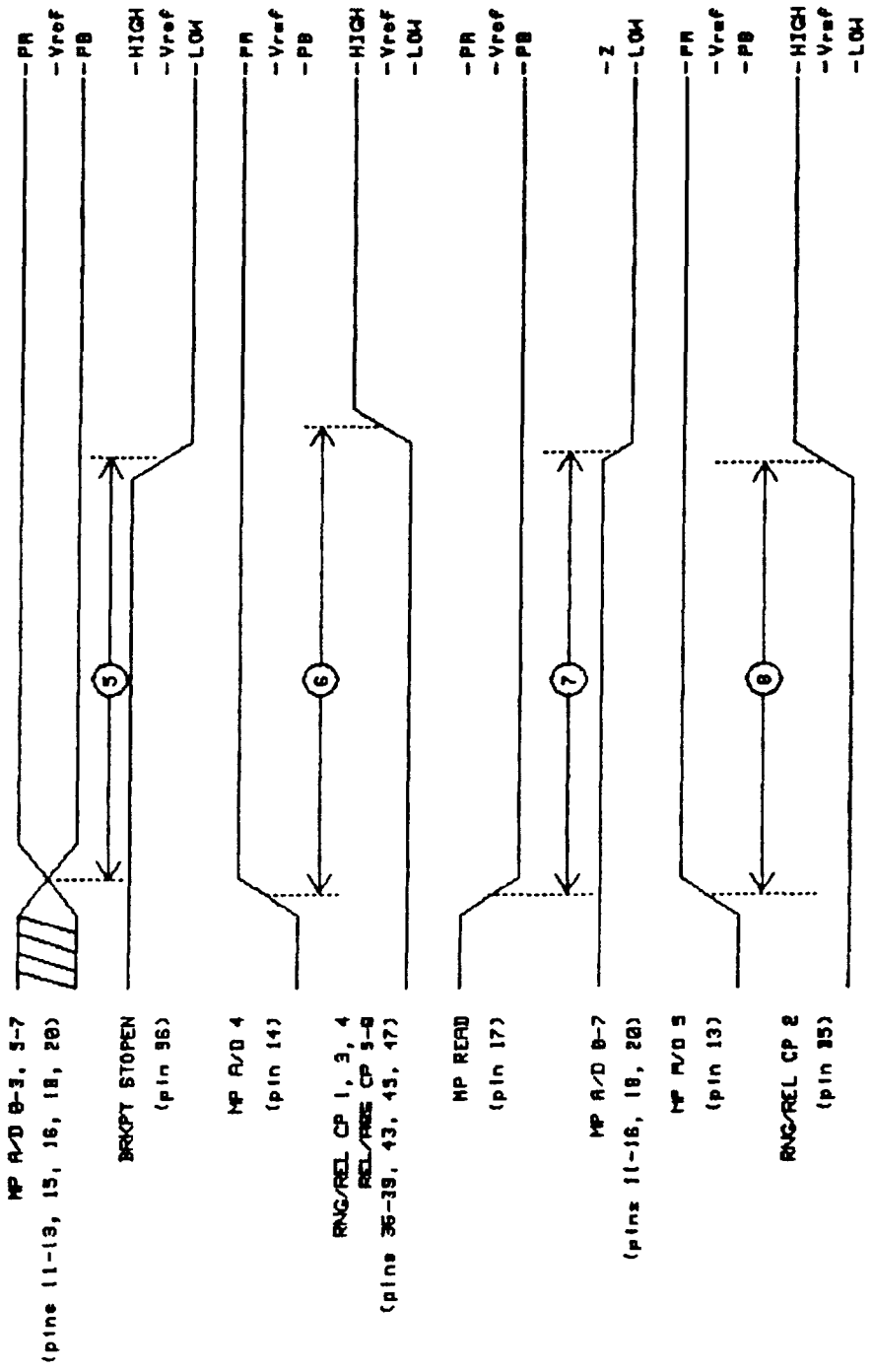


FIGURE 11b. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	CB ABS A12	2	2	50	pF	1-4
R2	CB ABS A14	3	2	75	pF	1-4
R3	CB ABS A11	4				
R4	CB ABS A13	5				
R5	CB ABS A10	6				
R6	CB ABS A17	7				
R7	CB ABS A16	8				
R8	CB ABS A15	9	2	75	pF	1-4
R9	MP READ	17	6	125	pF	1-4
R10	MP WRITE	19	6	125	pF	1-4
R11	MP ADRS 15	21	4	75	pF	1-4
R12	MP ADRS 14	22	4	75	pF	1-4
R13	MP ADRS 13	23	2	50	pF	1-4
R14	CB ABS A19	24	2	75	pF	1-4
R15	CB FCTN 1	26				
R16	CB ABS A21	27				
R17	CB FCTN 0	28				
R18	CB RD FCTN 2	29				
R19	CB FCTN 3	30				
R20	CB ABS A18	32	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	CB ABS A20	33	2	75	pF	1-4
R22	CB REL A14	34	2	75	pF	1-4
R23	CB REL A13	41	2	100	pF	1-4
R24	CB REL A12	42	2	75	pF	1-4
R25	PAGE SEL 0	44	2	100	pF	1-4
R26	CB REL A11	46	2	100	pF	1-4
R27	CB REL A15	48	2	125	pF	1-4
R28	CB REL A10	49	2	100	pF	1-4
R29	PAGE SEL 1	50	2	100	pF	1-4
R30	CB A/WR D3	54	2	50	pF	1-4
R31	CB A/WR D2	55				
R32	CB A/WR D1	56	2	50	pF	1-4
R33	CB A/WR D0	57	2	75	pF	1-4
R34	CB A/WR D7	58	2	75	pF	1-4
R35	MP ALE	59	6	175	pF	1-4
R36	CB A/WR D6	61	2	75	pF	1-4
R37	CB A/WR D4	62				
R38	CB OR	63	2	75	pF	1-4
R39	SPARE GRANT	64	2	150	pF	1-4
R40	I/O GRANT	65	2	100	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	ANY GRANT	66	4	150	pF	1-4
R42	MP ADRS 11	67	2	100	pF	1-4
R43	MP ADRS 10	68	2	125	pF	1-4
R44	MP ADRS 8	69	2	150	pF	1-4
R45	MP ADRS 9	70	2	125	pF	1-4
R46	CB A/WR D5	71	2	75	pF	1-4
R47	CP GRANT	72	2	125	pF	1-4
R48	BUS CLK	76	4	75	pF	1-4
R49	MP ADRS 15	77	4	75	pF	1-4
R50	CB A/WR D9	78	2	50	pF	1-4
R51	CB A/WR D8	79	2	50	pF	1-4
R52	PWR MCLR	80	6	217	pF	1-4
R53	MP ADRS 14	83	4	75	pF	1-4
R54	EX QUALIFIER	84	2	75	pF	1-4
R55	NDRO	85	2	75	pF	1-4
R56	MP ADRS 13	86	4	100	pF	1-4
R57	MP ADRS 12	87	2	75	pF	1-4
R58	FAN FAULT	88				
R59	OVERTEMP	89				
R60	LOGIC PFAULT	91	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R61	PRIMARY PWR	92	2	75	pF	1-4
R62	MAP 0	93				
R63	MAP 1	94	2	75	pF	1-4
R64	CB DATA T/O	97	4	75	pF	1-4
R65	CB ADRS T/O	98	4	75	pF	1-4
R66	BP DETECT EN	99	2	75	pF	1-4
	Bidirectionals					
R67	MP A/D 7	11	2	75	pF	1-4
R68	MP A/D 6	12				
R69	MP A/D 5	13				
R70	MP A/D 4	14				
R71	MP A/D 3	15				
R72	MP A/D 2	16				
R73	MP A/D 1	18				
R74	MP A/D 0	20	2	75	pF	1-4
R75	CB BUSY	52	4	75	pF	1-4
R76	CB ATR	53	8	200	pF	1-4
	Input					
R77	INSTRTN REF	82	4	100	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R77
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R77
Capacitance measurement	NA	± 2.4 pF	R1
	NA	± 3.65 pF	R2-R8
	NA	± 5.95 pF	R9,R10
	NA	± 3.55 pF	R11,R12
	NA	± 2.4 pF	R13
	NA	± 3.65 pF	R14-R22
	NA	± 4.9 pF	R23
	NA	± 3.65 pF	R24
	NA	± 4.9 pF	R25,R26
	NA	± 6.15 pF	R27
	NA	± 4.9 pF	R28,R29
	NA	± 2.4 pF	R30-R32
	NA	± 3.65 pF	R33,R34
	NA	± 8.45 pF	R35
	NA	± 3.65 pF	R36-R38
	NA	± 7.4 pF	R39
	NA	± 4.9 pF	R40
	NA	± 7.3 pF	R41
	NA	± 4.9 pF	R42
	NA	± 6.15 pF	R43
	NA	± 7.4 pF	R44
	NA	± 6.15 pF	R45
	NA	± 3.65 pF	R46
	NA	± 6.15 pF	R47
	NA	± 3.55 pF	R48,R49
	NA	± 2.4 pF	R50,R51
	NA	±10.5 pF	R52
	NA	± 3.55 pF	R53
	NA	± 3.65 pF	R54,R55
	NA	± 4.8 pF	R56
	NA	± 3.65 pF	R57-R63
	NA	± 3.55 pF	R64,R65

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
	NA	± 3.65 pF	R66-R74
	NA	± 3.55 pF	R75
	NA	± 9.6 pF	R76
	NA	± 4.8 pF	R77

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 12.

4. Abbreviations used are defined in 6.1.

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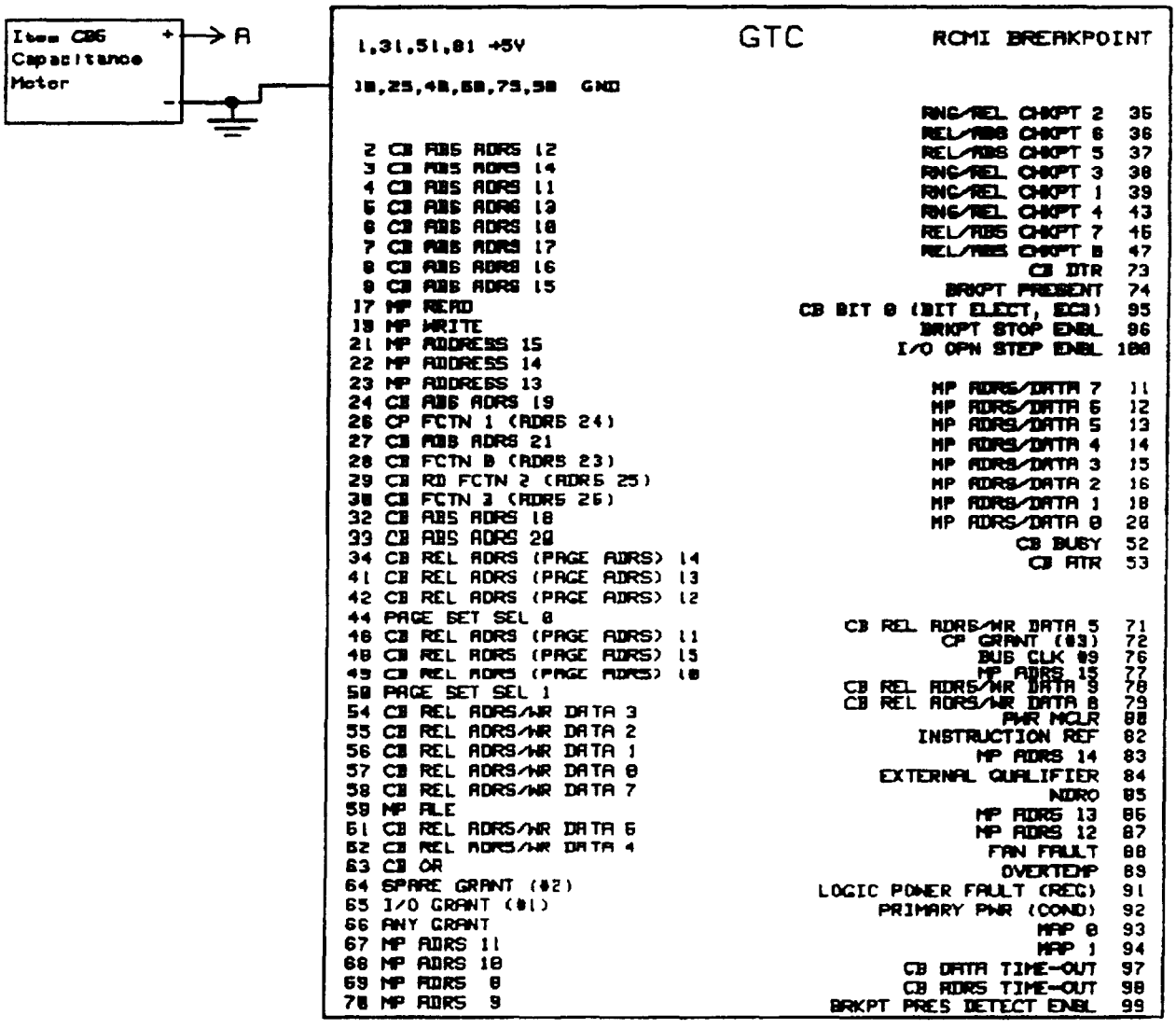


FIGURE 12. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T	PSTA	1	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T
Power supply current limit	2.45 A	NA	T
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T T
Power supply transient amplitude measurement	NA	± 20 mV	T

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 13 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 14.

4. Abbreviations used are defined in 6.1.

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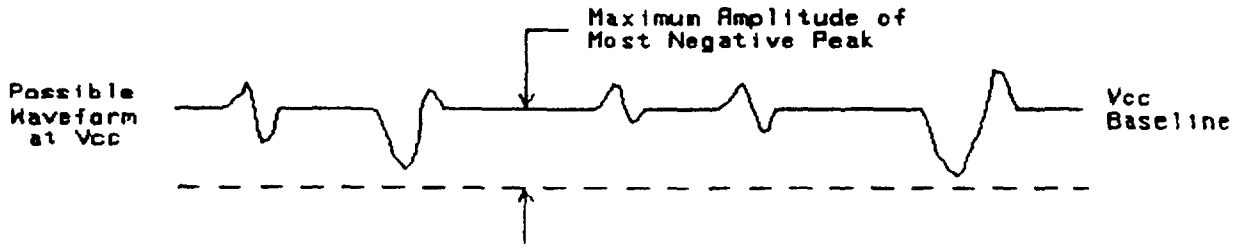
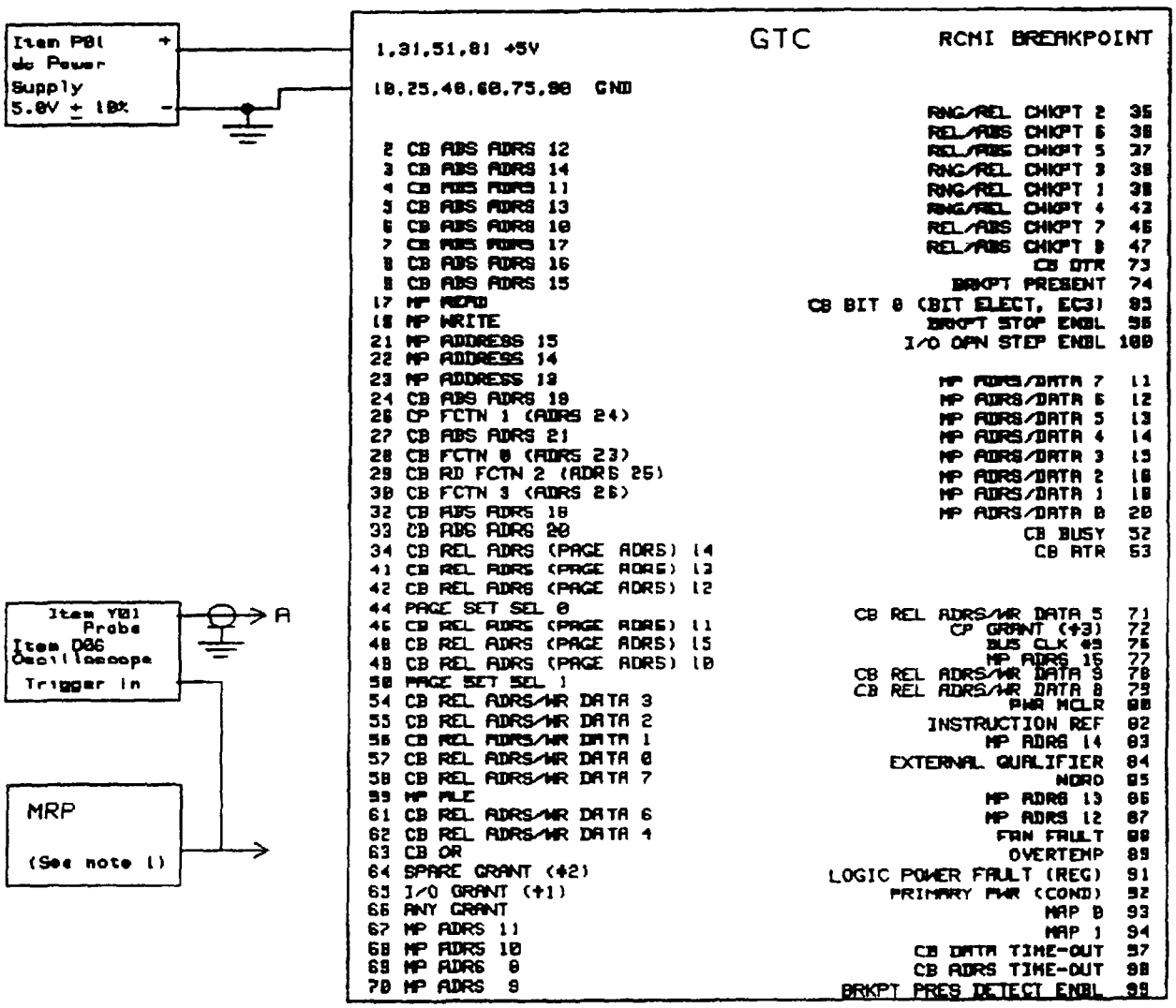


FIGURE 13. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 14. Power supply transient amplitude test circuit for table XIX.

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